Rate-Distortion-Function-Inspired CSI Feedback for FDD Massive MIMO-OFDM Communications

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***Abstract*—In this paper, we investigate the downlink channel state information (CSI) feedback for frequency-division duplexing (FDD) massive multiple-input multiple-output (MIMO) communications with orthogonal frequency division multiplexing (OFDM) modulation. By formulating CSI feedback as a lossy compression problem, we study its fundamental limit governed by the rate–distortion (R–D) function. To obtain a tractable surrogate for high-dimensional CSI, we leverage a variational upper bound that can be computed via variational autoencoder (VAE) training as a conservative benchmark. Building on this upper bound, we establish a set of sufficient optimality conditions that provide theoretical guidance for practical design. Guided by these conditions, we propose a VAE–based neural network, termed the Rate–Distortion-Function-Inspired Network (RDFI-Net), for CSI feedback. Specifically, RDFI-Net employs the hierarchical autoregressive probabilistic model to strengthen latent entropy modeling for efficient entropy coding, and incorporates enhanced neural modules to approximate the derived optimality conditions. Furthermore, we develop a low-complexity rate adaptation scheme that enables variable-rate CSI feedback without training and managing multiple models. Numerical results show that our proposed approach achieves superior performance.**

***Index Terms*—Massive MIMO, FDD, CSI feedback, rate-distortion theory, deep neural networks.**

I. Introduction

MASSIVE multiple-input multiple-output (MIMO) has become a cornerstone technology for wireless networks [1]. By deploying a large antenna array at a base station (BS), massive MIMO enables significant gains in spectral and energy efficiency. These performance improvements are critically reliant on having accurate and timely channel state information (CSI) at the transmitter. In frequency division duplex (FDD) systems, the lack of reciprocity between downlink and uplink channels necessitates that user equipment (UE) feedback downlink CSI to the BS [2]. However, massive MIMO in combination with orthogonal frequency division multiplexing (OFDM) leads to high-dimensional channel matrices due to the large number of antennas and subcarriers, imposing a substantial CSI feedback overhead. Therefore, reducing the feedback burden in massive MIMO-OFDM systems has been an indispensable research priority.

In recent years, deep learning (DL) based CSI feedback schemes have demonstrated superior performance attributed to the advanced feature extraction capabilities offered by deep neural networks (DNNs) [2]. These schemes commonly adopt a two-stage compression framework [2], [3], [4]: First, building on the discrete Fourier transform (DFT)-based channel model, the two-dimensional inverse discrete Fourier transform (2D-IDFT) is employed to transform the space-frequency domain channel matrix (SFDCM) to the DFT-based angular-delay domain channel matrix (ADCM). It is then truncated to reduce dimensionality and the resulting truncated ADCM (T-ADCM) is fed into DNNs for further compression, learning compact latent representations for feedback. At the BS, the T-ADCM is reconstructed from these compact representations, and finally SFDCM is recovered via zero-padding and 2D-DFT. In [4], the authors pioneer the use of deep autoencoders (AEs) for a secondary compression of the T-ADCM, inspiring subsequent advancements. In [5], variable-size convolutional kernels are designed to exploit the sparse feature inherent in ADCM. In [6], an attention mechanism is employed in a lightweight network to enhance CSI feedback performance. In [7], a full attention network based on Transformer architecture is proposed. In [8], the authors propose a sparse autoencoder–based scheme to improve both feedback efficiency and robustness. In [9], JC-ResNet is proposed to enhance CSI feedback by improving recovery performance under limited-bit quantization. In addition, a denoising neural network named AnciNet is developed in [10], which allows for the channel estimation errors. Moreover, to facilitate lightweight deployment, ENet is developed [11] by leveraging the inherent correlation characteristics of channel matrices. The aforementioned works mainly treat CSI as unstructured data and achieve compression based on dimension reduction, whereas subsequent studies have incorporated entropy coding to enhance feedback efficiency [12], [13], [14], [15]. By

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mapping the high-dimensional CSI into a latent space and quantizing the resulting latent representations into discrete symbols, these methods adopt entropy coding based on learned probability models. The authors in [12] pioneered the integration of quantization and entropy coding into an autoencoder and used a weighted rate–distortion loss function to train the neural network. Motivated by this work, single-layer VAE-based neural architectures were introduced in [13] and [14]. In [15], the authors propose a sparse VAE-based framework that jointly compresses a sparse latent representation and its binary support, and incorporate sparsity-inducing regularization into the training objective.

In practice, deploying DL-based CSI feedback also faces two recurring bottlenecks: (i) the high cost of collecting representative CSI datasets, and (ii) performance degradation under distribution shifts between the training and deployment environments [16], [17]. For instance, the authors in [17] investigate environment mismatch and improve robustness by explicitly modeling the CSI distribution shift across different environments and designing an environment-generalizable learning framework. Meanwhile, the authors in [16] emphasize the data burden in realistic deployment and propose a knowledge-driven meta-learning strategy, enabling fast adaptation to a new scenario with limited target retraining data. The trade-off between feedback overhead and reconstruction accuracy in the aforementioned works is mainly fixed by either the latent dimension or the weighting term. Accordingly, to accommodate dynamic feedback bandwidth constraints, several variable-rate CSI feedback schemes have been investigated [18], [19], [20], [21], [22]. For example, the authors in [18] propose a multi-decoder design at the BS that supports four predefined feedback-overhead options. Likewise, the authors in [19] develop a serial compression framework that enables up to four selectable overhead configurations. In [20], the authors propose a VQ-VAE–based CSI feedback scheme that supports variable-rate feedback over a predefined rate set without deploying multiple models. However, most existing variable-rate mechanisms either require maintaining multiple models or restrict rate adaptation to a predefined set of operating points.

Despite the significant progress in the research area of DL-based CSI feedback, existing works generally assume that the complete downlink CSI can be reconstructed from the T-ADCM with negligible error at the BS [2], [3], [4], [5]. However, significant non-zero elements are spread into the tail columns of the DFT-based ADCM due to spectral leakage. Consequently, simply truncating these columns leads to an inherent reconstruction error. While the authors in [23] recognize the stripe patterns caused by spectral leakage and exploit them as intrinsic features to improve feedback performance, the performance degradation due to truncation remains unaddressed. From a rate-control perspective, although [21], [22] enable fine-grained rate adaptation with a single model without restricting operation to a predefined set of operating points, how to incorporate such fine-grained rate control into entropy-coded DL-based CSI feedback schemes remains underexplored.

More critically, most existing DL-based CSI feedback schemes primarily focus on neural architecture and module design, and only a few explicitly connect their designs to a Shannon-theoretic rate–distortion (R–D) viewpoint. This connection is crucial because, once CSI feedback is cast as lossy source coding over a digital feedback bit-pipe, its fundamental limit—the minimum feedback rate required to achieve a target distortion—is characterized by the R–D function. Several recent DL-based CSI feedback studies incorporate quantization and entropy coding and optimize a weighted rate–distortion objective [12], [13], [14], [15]. By explicitly learning a probability model for the latent symbols, these approaches have been shown to be more efficient than the dimension-reduction autoencoder pipelines. However, although these entropy-coded designs reflect an operational rate–distortion trade-off and thus bear a clear connection to R–D theory, to the best of our knowledge there is still a lack of work that explicitly provides a principled and computable (approximate) characterization of the Shannon R–D limit for CSI feedback.

In this paper, we investigate the downlink CSI feedback for FDD massive MIMO-OFDM communications. First, leveraging the refined beam based channel model [25], [26], we adopt the refined beam domain channel matrix for feedback, thereby mitigating the truncation-induced error. Moreover, we study the fundamental R–D limit of CSI feedback through a computable variational upper bound that serves as a benchmark for practical schemes. The main contributions are summarized as follows.

- We study CSI feedback through the scheme-independent Shannon R–D limit. Since this limit is generally intractable for high-dimensional CSI sources with unknown distributions, we leverage a decoder-induced variational upper bound that can be evaluated through VAE-based optimization. Analysis of this upper bound further yields a set of ideal sufficient conditions that provide guidance for practical neural codec design.
- Guided by the optimality conditions, we propose a VAE-based neural codec for CSI feedback, termed RDFI-Net, which translates these conditions into implementable design principles. In contrast to existing VAE-based CSI feedback schemes that typically adopt single-layer latent models, RDFI-Net introduces a hierarchical autoregressive probabilistic model (HAPM) with explicit inter-layer dependencies, providing a stronger entropy model and more accurate symbol probabilities for entropy coding.
- Furthermore, we develop a low-complexity rate adaptation scheme that avoids training and managing multiple models. In particular, we propose a variable-rate training strategy and achieve rate adaptation through a learnable Adaptive Quantization Factor (AQF), which governs the quantization bin size to balance the feedback overhead and reconstruction accuracy.

To make our contributions more explicit, Table I positions this paper relative to existing works on DL-based CSI feedback and highlights the key differences.

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TABLE I
COMPARISON OF OUR WORK WITH REPRESENTATIVE EXISTING CSI FEEDBACK STUDIES. DETAILED DISCUSSION IS PROVIDED IN SECTIONS I, III, AND IV

Research	Computable R–D Upper-Bound Benchmark	Entropy Coding	Probabilistic Model	Variable-Rate Mechanism
[4], [5], [6], [7]	×	×	×	×
[18], [19]	×	×	×	Multiple Models, Predefined Rates
[21], [22]	×	×	×	Single Model, Fine-Grained
[12]	×	✓	Single Layer	Multiple Models, Predefined Rates
[13], [14], [24]	×	✓	Single Layer	×
[15]	×	✓	Sparse Two-Tier	×
[20]	×	×	×	Single Model, Predefined Rates
Ours	✓	✓	**Hierarchical Autoregressive**	**Single Model, Fine-Grained**

The rest of this paper is organized as follows. In Section II, we present the system configurations, the refined beam based channel model and the system model of CSI feedback utilizing the refined beam domain channel matrix. In Section III, we study the fundamental R–D limit of CSI feedback and develop a tractable variational upper-bounding characterization together with ideal sufficient conditions. In Section IV, we detail the proposed RDFI-Net for CSI feedback. Simulation results are provided in Section V, followed by a discussion in Section VI. The conclusion is given in Section VII.

Notations: Throughout this paper, scalar variables are denoted by regular letters, while column vectors and matrices are represented by boldface lowercase and uppercase letters, respectively. The set of positive integers up to N is defined as $\mathcal{Z}_N^+ \triangleq \{1,\ldots,N\}$. The operator $\mathrm{Re}(\cdot)$ denotes the operation of taking the real part of a complex number, vector, or matrix. $*$ denotes the convolution operation. $\mathrm{vec}(\cdot)$ is defined as the vectorization operation. The expectation operator is denoted by $\mathbb{E}[\cdot]$. Unless otherwise specified, $\log(\cdot)$ denotes the base-2 logarithm, $\ln(\cdot)$ denotes the natural logarithm, and $D_{\mathrm{KL}}(\cdot \parallel \cdot)$ is measured in bits. The complex matrix space of dimension $M \times N$ is represented by $\mathbb{C}^{M\times N}$. The Kronecker product is denoted by $\otimes$. The $N \times N$ identity matrix is denoted by $\mathbf{I}_N$, and $\mathbf{I}_{N,M}$ represents $[\mathbf{I}_N\ \mathbf{0}_{N,M-N}]$ when $N < M$ or $[\mathbf{I}_M\ \mathbf{0}_{M,N-M}]^T$ when $N > M$. The (i,j)-th entry of a matrix $\mathbf{A}$ is denoted by $[\mathbf{A}]_{i,j}$. An N-dimensional normalized DFT matrix is denoted by $\mathbf{F}_N$. The indicator function, denoted by $g_{a,b}(\cdot)$, takes the value 1 if its argument falls within the interval (a,b), and 0 otherwise. The uniform distribution is denoted by $\mathcal{U}(a,b)$. The Gaussian distribution with mean μ and variance σ^2 is denoted by $\mathcal{N}(\mu,\sigma^2)$, and its probability density function (PDF) is denoted by $f_{\mu,\sigma^2}(\cdot)$. $\mathcal{Q}(\cdot)$ denotes the scalar round quantization operation.

II. SYSTEM MODEL AND CHANNEL MODEL

In this section, we first present the system configurations and introduce the refined beam based channel model. Then, the system model of downlink CSI feedback with the refined beam domain channel matrix is proposed.

A. System Configuration

Consider a single-cell massive MIMO-OFDM system[1] operating in FDD mode, where the BS equipped with a uniform linear array (ULA) of N_t antennas communicates with K single-antenna user equipments (UEs). The carrier frequency is denoted as f_c. The antennas are spaced $0.5\lambda_c$, where $\lambda_c \triangleq c/f_c$ denotes the wavelength, and c denotes the speed of light. The total number of subcarriers is denoted as N_c, among which N_p subcarriers are allocated for downlink pilot signal transmission. The subcarrier spacing is $\Delta_f = \frac{1}{N_c T_s}$, where T_s is defined as the sampling interval length. The length of the cyclic prefix (CP) is denoted as N_g. Moreover, we assume the channel is quasi-static. In this paper, we focus on CSI feedback transmitted over a dedicated uplink. The main notations used in this section are listed in Table II.

TABLE II
SYMBOL LIST ON SYSTEM CONFIGURATIONS

Notation	Definition
K	Number of UEs
N_t	Number of BS antennas
f_c	Carrier frequency
c	Speed of light
λ_c	Wavelength, $\lambda_c \triangleq c/f_c$
N_c	Total number of OFDM subcarriers
N_p	Number of effective subcarriers
Δf	Subcarrier spacing, $\Delta f \triangleq \frac{1}{N_c T_s}$
T_s	Sampling interval length
N_g	Cyclic prefix (CP) length
N_1	Start index of pilot subcarriers, $N_1 \triangleq -\frac{N_p}{2}+1$
N_2	End index of pilot subcarriers, $N_2 \triangleq \frac{N_p}{2}$

B. Refined Beam-Based Channel Model

Define the spatial domain steering vector $\mathbf{v}(u)$ and the frequency domain steering vector $\mathbf{u}(\tau)$ as

$$\mathbf{v}(u) = [1, e^{-j\pi u} \cdots e^{-j\pi(N_t-1)u}]^T \in \mathbb{C}^{N_t\times 1}, \tag{1a}$$

$$\mathbf{u}(\tau) = \left[e^{-j2\pi\Delta_f N_1 \tau} \cdots e^{-j2\pi\Delta_f N_2 \tau}\right]^T \in \mathbb{C}^{N_p\times 1}, \tag{1b}$$

where $N_1 \triangleq -\frac{N_p}{2}+1$ and $N_2 \triangleq \frac{N_p}{2}$ denote the start and end indices of the subcarriers allocated for downlink channel estimation, respectively. The downlink SFDCM between the BS and the k-th UE is given by

$$\mathbf{H}_k^{\mathrm{SF}} = \sum_{p=1}^{P_k} h_{k,p}\mathbf{v}(u_{k,p})\,\mathbf{u}^T(\tau_{k,p}), \tag{2}$$

where P_k is the number of propagation paths associated with the k-th user, $u_{k,p} \triangleq \cos\theta_{k,p}$ denotes the directional cosine corresponding to the angle of departure $\theta_{k,p}$, $\tau_{k,p}$ denotes the

[1]We consider massive MIMO-OFDM as it is a standard and practically relevant broadband setting in which the CSI feedback overhead scales with both the number of BS antennas and the number of subcarriers, resulting in high-dimensional CSI and a significant feedback burden.

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time delay of the p-th path, and $h_{k,p} \triangleq \alpha_{k,p} e^{-j2\pi f_c \tau_{k,p}}$, with $\alpha_{k,p}$ denoting the complex path gain.

The parameters $(u_{k,p}, \tau_{k,p})$ of each path for each UE are restricted to the ranges $\mathcal{B}_u \triangleq [-1, 1]$ and $\mathcal{B}_\tau \triangleq [0, N_g T_s)$, respectively. For subsequent processing, we uniformly divide these ranges into multiple disjoint subsets as

$$\mathcal{B}_u = \bigcup_{i=1}^{N_{\text{an}}} \mathcal{B}_{u,i}, \quad \mathcal{B}_{u,i} = [u_i, u_{i+1}), \tag{3a}$$

$$\mathcal{B}_\tau = \bigcup_{j=1}^{N_{\text{de}}} \mathcal{B}_{\tau,j}, \quad \mathcal{B}_{\tau,j} = [\tau_j, \tau_{j+1}) \tag{3b}$$

where u_i, τ_j are the sampled directional cosines and delays, respectively, with $u_i = \frac{2(i-1)-N_{\text{an}}}{N_{\text{an}}}$, $\tau_j = \frac{(j-1)N_f}{N_{\text{de}} N_p \Delta f}$. Here, $N_f \triangleq \lceil \frac{N_p N_g}{N_c} \rceil$, $N_{\text{an}} = F_{\text{an}} N_t$, and $N_{\text{de}} = F_{\text{de}} N_f$ where F_{an} and F_{de} are referred to as fine factors. Denoting $\Gamma_k^{\text{SF}} \triangleq \{(u_{k,1}, \tau_{k,1}), \cdots, (u_{k,P_k}, \tau_{k,P_k})\}$ as the path parameter set of the k-th UE and let $\mathcal{B}_{i,j} \triangleq \mathcal{B}_{u,i} \times \mathcal{B}_{\tau,j}$, the SFDCM in (2) can be rewritten as

$$\mathbf{H}_k^{\text{SF}} = \sum_{i,j} \sum_{(u_{k,p}, \tau_{k,p}) \in \Gamma_k^{\text{SF}} \cap \mathcal{B}_{i,j}} h_{k,p} \mathbf{v}(u_{k,p}) \mathbf{u}^T(\tau_{k,p}). \tag{4}$$

When N_{an} and N_{de} are sufficiently large, the steering vectors $\mathbf{v}(u)$ in $\mathcal{B}_{u,i}$ and $\mathbf{u}(\tau)$ in $\mathcal{B}_{\tau,j}$ can be well approximated by the sampled spatial steering vectors $\mathbf{v}(u_i)$ and the sampled frequency steering vectors $\mathbf{u}(\tau_j)$, respectively [25], [26]. Consequently, (4) can be reformulated as

$$\mathbf{H}_k^{\text{SF}} \approx \sum_{i,j} \tilde{h}_k(u_i, \tau_j) \mathbf{v}(u_i) \mathbf{u}^T(\tau_j), \tag{5}$$

$$\tilde{h}_k(u_i, \tau_j) = \sum_{(u_{k,p}, \tau_{k,p}) \in \Gamma_k^{\text{SF}} \cap \mathcal{B}_{i,j}} h_{k,p}. \tag{6}$$

Defining $\mathbf{V} = [\mathbf{v}(u_1), \ldots, \mathbf{v}(u_{N_{\text{an}}})] \in \mathbb{C}^{N_t \times N_{\text{an}}}$ and $\mathbf{U} = [\mathbf{u}(\tau_1), \ldots, \mathbf{u}(\tau_{N_{\text{de}}})] \in \mathbb{C}^{N_p \times N_{\text{de}}}$, the SFDCM in (5) then becomes

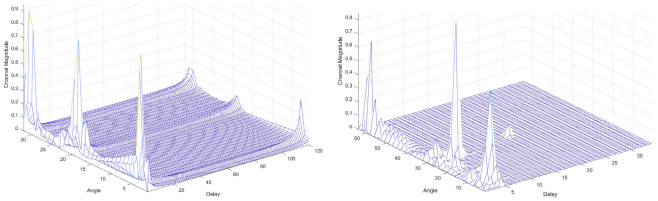
$$\mathbf{H}_k^{\text{SF}} \approx \mathbf{V} \mathbf{H}_k^{\text{BD}} \mathbf{U}^T, \tag{7}$$

where $\mathbf{H}_k^{\text{BD}} \in \mathbb{C}^{N_{\text{an}} \times N_{\text{de}}}$ is referred to as the refined beam domain channel matrix (BDCM), and (7) as the refined beam based channel model. The refined beam based channel model in (7) generalizes the widely adopted DFT-based channel model used in existing CSI feedback studies, with the latter as a special case when the fine factors are set to one. Specifically, this yields

$$\mathbf{H}_k^{\text{SF}} \approx \mathbf{F}_{N_t} \tilde{\mathbf{H}}_k^{\text{AD}} \mathbf{F}_{N_p \times N_f}^H, \tag{8}$$

where $\mathbf{F}_{N_p \times N_f} \triangleq \mathbf{F}_{N_p} \mathbf{I}_{N_p, N_f}$, and $\tilde{\mathbf{H}}_k^{\text{AD}} \triangleq \mathbf{H}_k^{\text{AD}} \mathbf{I}_{N_p, N_f}$ denotes the T-ADCM with $\mathbf{H}_k^{\text{AD}} = \mathbf{F}_{N_t}^H \mathbf{H}_k^{\text{SF}} \mathbf{F}_{N_p}$ representing the DFT-based ADCM.

Note that the truncation scheme widely employed in existing works is built upon (8), which assumes that only the first N_f columns of $\mathbf{H}_k^{\text{AD}}$ contain significant coefficients. However, this model is accurate only when the number of effective subcarriers is sufficiently large. In practice, spectral leakage causes significant coefficients of the DFT-based ADCM to spread into the tail columns, as illustrated in Fig. 1a and explained through a continuous-window interpretation in Appendix A, leading to non-negligible reconstruction errors.



(a) DFT-based ADCM. (b) BDCM ($F_{\text{an}} = F_{\text{de}} = 2$).

Fig. 1. Comparison of DFT-based ADCM and BDCM.

By setting the fine factors greater than one, the BDCM extends the DFT-based ADCM to the refined beam domain, thereby achieving higher angular–delay resolution and mitigating spectral leakage, as shown in Fig. 1b. Throughout this paper, we set the fine factors $F_{\text{an}}, F_{\text{de}} > 1$ and, with a slight abuse of terminology, omit the qualifier "refined" thereafter.

C. Downlink CSI Feedback With BDCM

Given the notable advantages offered by BDCM, we adopt it for feedback, in contrast to existing methods that utilize T-ADCM.

For convenience, we assume that each UE can estimate the beam domain channel matrix $\mathbf{H}_k^{\text{BD}}$ from the downlink pilot signals through beam domain channel estimation algorithms [25], [26], as detailed in Appendix B. Subsequently, UE encodes $\mathbf{H}_k^{\text{BD}}$ into codewords $\mathbf{v}_k$, which is expressed as

$$\mathbf{v}_k = f_{\text{en}}\left(\mathbf{H}_k^{\text{BD}}\right), \tag{9}$$

where $f_{\text{en}}(\cdot)$ denotes the encoding function of the encoder. After $\mathbf{v}_k$ is transmitted back to the BS, $\hat{\mathbf{H}}_k^{\text{BD}}$ is recovered by the decoder, which is expressed as

$$\hat{\mathbf{H}}_k^{\text{BD}} = f_{\text{de}}(\mathbf{v}_k), \tag{10}$$

where $f_{\text{de}}(\cdot)$ denotes the decoding function of the decoder. Finally, $\hat{\mathbf{H}}_k^{\text{SF}}$ is recovered from $\hat{\mathbf{H}}_k^{\text{BD}}$ as

$$\hat{\mathbf{H}}_k^{\text{SF}} = \mathbf{V} \hat{\mathbf{H}}_k^{\text{BD}} \mathbf{U}^T. \tag{11}$$

Importantly, BDCM typically achieves a near-lossless representation of the SFDCM when the fine factors are set to 2 [25]. Under this configuration, $\mathbf{H}_k^{\text{BD}}$ yields a compact yet accurate representation of the channel, preserving the essential characteristics and retaining critical features that are often lost in T-ADCM.

Remark 1: In practice, the BDCM used as the input of (9) is obtained at the UE from noisy downlink pilots via a beam-domain estimator. Let $\mathbf{H}_{k,0}^{\text{BD}}$ denote the underlying (noise-free) BDCM associated with the true channel, and let $\mathbf{H}_k^{\text{BD}}$ denote the UE-available estimate used in (9). Then the BS-side end-to-end error admits

$$\hat{\mathbf{H}}_k^{\text{BD}} - \mathbf{H}_{k,0}^{\text{BD}} = \underbrace{\left(\hat{\mathbf{H}}_k^{\text{BD}} - \mathbf{H}_k^{\text{BD}}\right)}_{\text{feedback error}} + \underbrace{\left(\mathbf{H}_k^{\text{BD}} - \mathbf{H}_{k,0}^{\text{BD}}\right)}_{\text{estimation error}}.$$

Following the common CSI feedback problem formulation [2], we treat downlink training and channel estimation as an

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upstream block and focus on the subsequent CSI feedback stage given the UE-available estimate $\mathbf{H}_k^{\mathrm{BD}}$. A joint end-to-end treatment that simultaneously considers downlink channel estimation and feedback (e.g., under a JSCC framework) can be found in [27].

III. Fundamental Limit of CSI Feedback: Upper-Bounding Characterization and Optimality Conditions

In this section, we first study the fundamental limit of CSI feedback specified by the R–D function by formulating CSI feedback as a lossy compression problem. Subsequently, we evaluate a tractable variational upper bound via VAEs, and then establish a set of ideal sufficient conditions that guide practical codec design.[2]

A. Problem Formulation

By formulating CSI feedback as a lossy compression problem, our analysis establishes a general framework that is independent of specific channel representations. Accordingly, we employ the generic notation $\mathbf{H}$ for the channel matrix and defer the introduction of domain-specific indices until required by the analysis. We denote $\mathcal{H}$ as the space of $\mathbf{H}$ and $\hat{\mathcal{H}}$ as the reproduction space of $\hat{\mathbf{H}}$.

Under the mild assumption that UEs feed back downlink CSI to the BS independently and the feedback codewords are transmitted without error, the feedback process can be modeled as a memoryless high-dimensional information source. As a consequence, the minimum feedback overhead required to achieve a distortion no more than D is determined by the R-D function [28], [29], which is given by

$$R(D) = \inf_{Q_{\hat{\mathbf{H}}|\mathbf{H}}:\mathbb{E}[\rho(\mathbf{H},\hat{\mathbf{H}})]\le D} I(\mathbf{H};\hat{\mathbf{H}}), \tag{12}$$

where $\rho : \mathcal{H}\times\hat{\mathcal{H}} \to [0,\infty)$ denotes the distortion metric and $Q_{\hat{\mathbf{H}}|\mathbf{H}}$ denotes the transition kernel from $\mathcal{H}$ to $\hat{\mathcal{H}}$.

The R–D function in (12) establishes a fundamental lower bound: no coding scheme can achieve distortion D with a rate below $R(D)$. Consequently, characterization of the R-D function is critical. However, analytical expressions for this function generally remain elusive, and the specific calculation for the high-dimensional CSI source presents particularly formidable challenges.

B. Upper-Bounding R-D Function With VAEs

Since the true R–D function $R(D)$ is generally intractable to be evaluated exactly for high-dimensional CSI, we leverage a computable variational upper bound [30], [31] as a tractable surrogate to provide an upper-bounding characterization. This bound is mathematically rigorous and can achieve equality under some ideal conditions.

Let $\mathcal{Z}$ be a latent space and $\omega : \mathcal{Z} \to \hat{\mathcal{H}}$ be a mapping from the latent space to the reproduced space. This induces a new compression problem $\mathcal{H} \to \mathcal{Z} \to \hat{\mathcal{H}}$ under a new distortion function $\rho_\omega : \mathcal{H}\times\mathcal{Z} \to [0,\infty)$, which is given by

$$\rho_\omega(\mathbf{H},\mathbf{Z}) = \rho(\mathbf{H},\omega(\mathbf{Z})), \tag{13}$$

where ρ denotes the distortion metric defined in (12). Furthermore, we define the decoder-induced R-D function corresponding to ω as

$$R_\omega(D) = \inf_{Q_{\mathbf{Z}|\mathbf{H}}:\mathbb{E}[\rho(\mathbf{H},\omega(\mathbf{Z}))]\le D} I(\mathbf{H};\mathbf{Z}), \tag{14}$$

where $Q_{\mathbf{Z}|\mathbf{H}}$ is referred to as the transition kernel from $\mathcal{H}$ to $\mathcal{Z}$. $R_\omega(D)$ is a mathematically rigorous approximation of $R(D)$ in (12), as stated in the following lemma.

Lemma 1 (cf. [30]): $R_\omega(D)$ constitutes an upper bound on (12), i.e., $R_\omega(D) \ge R(D)$ for any $D \ge 0$. Equality holds at distortion D if there exists an $R(D)$-optimal reproduction $\hat{\mathbf{H}}$ with a latent representation $\mathbf{Z}$ such that $\hat{\mathbf{H}} = \omega(\mathbf{Z})$ and $I(\mathbf{H};\mathbf{Z}\mid\hat{\mathbf{H}}) = 0$. In particular, this condition is guaranteed when the decoder ω is bijective.

From the lemma above, $R_\omega(D)$ provides an upper-bounding characterization of $R(D)$. We further reformulate (14) as

$$\begin{aligned} R_\omega(D) &= \inf_{P_{\mathbf{Z}}} \inf_{Q_{\mathbf{Z}|\mathbf{H}}} D_{\mathrm{KL}}(P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}} \,\|\, P_{\mathbf{H}}P_{\mathbf{Z}}) \\ &= \inf_{P_{\mathbf{Z}}} \inf_{Q_{\mathbf{Z}|\mathbf{H}}} \mathbb{E}_{\mathbf{H}\sim P_{\mathbf{H}}}\left[D_{\mathrm{KL}}(Q_{\mathbf{Z}|\mathbf{H}} \,\|\, P_{\mathbf{Z}})\right] \\ &\quad \text{s.t. } \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\left[\rho_\omega(\mathbf{H},\mathbf{Z})\right] \le D, \end{aligned} \tag{15}$$

where $P_{\mathbf{Z}}$ is referred to as the prior distribution of $\mathbf{Z}$ and $Q_{\mathbf{Z}|\mathbf{H}}$ is defined as in (14). Optimization of (15) is equivalent to an unconstrained optimization problem, as stated in the following proposition.

Proposition 1: For a fixed Lagrangian multiplier $\lambda > 0$, R-D pair $(\mathcal{R}_\lambda^*, \mathcal{D}_\lambda^*)$ lies on $R_\omega(D)$, where

$$\mathcal{R}_\lambda^* = \mathbb{E}_{\mathbf{H}\sim P_{\mathbf{H}}}\left[D_{\mathrm{KL}}(Q_{\mathbf{Z}|\mathbf{H}}^{\lambda*} \,\|\, P_{\mathbf{Z}}^{\lambda*})\right], \tag{16a}$$

$$\mathcal{D}_\lambda^* = \mathbb{E}_{P_{\mathbf{H}}Q_{\hat{\mathbf{Z}}|\mathbf{H}}^{\lambda*}}\left[\rho_\omega(\mathbf{H},\mathbf{Z})\right]. \tag{16b}$$

Here $\left(Q_{\mathbf{Z}|\mathbf{H}}^{\lambda*}, P_{\mathbf{Z}}^{\lambda*}\right)$ is the solution of the following optimization problem

$$\left(Q_{\mathbf{Z}|\mathbf{H}}^{\lambda*}, P_{\mathbf{Z}}^{\lambda*}\right) = \arg\min_{Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}} \mathcal{L}(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}; \lambda), \tag{17}$$

where the objective function is expressed as

$$\begin{aligned} \mathcal{L}(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}; \lambda) &= \mathbb{E}_{P_{\mathbf{H}}}\left[D_{\mathrm{KL}}(Q_{\mathbf{Z}|\mathbf{H}} \,\|\, P_{\mathbf{Z}})\right] \\ &\quad + \lambda\mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\left[\rho_\omega(\mathbf{H},\mathbf{Z})\right]. \end{aligned} \tag{18}$$

Proof: Please see in Appendix C. ■

Remark 2: While the objective function in (18) differs from the standard Lagrangian form, minimizing it for a given $\lambda > 0$ is algebraically equivalent to applying the Lagrangian multiplier method to solve the following constrained optimization problem:

$$\inf_{Q_{\mathbf{Z}|\mathbf{H}}} I(\mathbf{H};\mathbf{Z}), \quad \text{s.t.} \quad \mathbb{E}[\rho_\omega(\mathbf{H},\mathbf{Z})] \le D_\lambda^*, \tag{19}$$

where D_λ^*, as defined in (16b), denotes the distortion achieved by a selected minimizer $Q_{\mathbf{Z}|\mathbf{H}}^{\lambda*}$ for the given λ. Since $R_\omega(D)$ is convex, the corresponding solution identifies a point on its

[2]Compared with [27], which jointly models downlink training and uplink feedback to study end-to-end CSIT acquisition, we adopt the standard CSI-feedback abstraction and focus on compression of an estimated CSI under a reliable digital-feedback assumption.

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boundary. Similar formulations have been adopted in [29] and [32].

Although $R_\omega(D)$ in (14) can be reformulated as an unconstrained optimization problem with the objective given in (18), directly solving this problem is challenging because the optimization variables are probability distributions. Nevertheless, by parameterizing $Q_{\mathbf{Z}|\mathbf{H}}$ and $P_{\mathbf{Z}}$ with DNNs, the variational objective can be optimized using a VAE-style encoder–decoder architecture, as stated below.

Proposition 2: For a fixed $\lambda > 0$, suppose that $Q_{\mathbf{Z}|\mathbf{H}}$, $P_{\mathbf{Z}}$, and the decoder ω are parameterized by DNNs. If the likelihood $P_{\mathbf{H}|\mathbf{Z}}$ is chosen such that $-\log P_{\mathbf{H}|\mathbf{Z}} = \lambda\rho_\omega(\mathbf{H},\mathbf{Z}) + C_\lambda$, where C_λ is independent of the trainable parameters, then minimizing (18) is equivalent, up to an additive constant, to VAE-style training.

Proof: The negative evidence lower bound of a VAE [30], [33] is

$$\mathcal{L}_{\mathrm{VAE}} = \mathbb{E}_{P_{\mathbf{H}}}\left[D_{\mathrm{KL}}\left(Q_{\mathbf{Z}|\mathbf{H}} \parallel P_{\mathbf{Z}}\right)\right] - \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\left[\log P_{\mathbf{H}|\mathbf{Z}}\right]. \tag{20}$$

Using $-\log P_{\mathbf{H}|\mathbf{Z}} = \lambda\rho_\omega(\mathbf{H},\mathbf{Z}) + C_\lambda$, we obtain $\mathcal{L}_{\mathrm{VAE}} = \mathcal{L}\left(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}; \lambda\right) + C_\lambda$. Therefore, the two objectives have the same minimizers. ∎

Remark 3: The "transition kernel" $Q_{\mathbf{Z}|\mathbf{H}}$ defined in (14) is also referred to as "posterior distribution" in the context of VAEs. For terminological consistency throughout this paper, we will consistently refer to $Q_{\mathbf{Z}|\mathbf{H}}$ as "transition kernel".

C. Optimality Conditions for CSI Feedback

Building on the discussion above, we now establish a set of sufficient optimality conditions for CSI feedback, which are formally stated in the following proposition.

Proposition 3: Sufficient conditions for achieving the optimal CSI feedback are as follows:

1) At D^*_λ, an $R(D^*_\lambda)$-optimal reproduction $\hat{\mathbf{H}}$ can be expressed as $\hat{\mathbf{H}} = \omega(\mathbf{Z})$ with $I(\mathbf{H};\mathbf{Z} \mid \hat{\mathbf{H}}) = 0$; a bijective ω is a sufficient special case.
2) The optimization in (17) attains its global optimum.
3) The coding scheme achieves, for any $\mathbf{H}_i$, a conditional expected ideal codelength equal to $D_{\mathrm{KL}}\left(Q^{\lambda*}_{\mathbf{Z}|\mathbf{H}_i} \parallel P^{\lambda*}_{\mathbf{Z}}\right)$, where $Q^{\lambda*}_{\mathbf{Z}|\mathbf{H}}$ and $P^{\lambda*}_{\mathbf{Z}}$ are defined in (17).

Proof: If Condition 1) holds, $R_\omega(D^*_\lambda) = R(D^*_\lambda)$, so that the upper bound attains equality at the considered operating point. If Conditions 2) and 3) also hold, the average feedback overhead equals $\mathcal{R}^*_\lambda$ with the average distortion $\mathcal{D}^*_\lambda$. Hence, the R–D pair $(\mathcal{R}^*_\lambda, \mathcal{D}^*_\lambda)$ lies on $R(D)$, which implies optimality. ∎

Although the sufficient conditions in Proposition 3 are idealized, satisfying them exactly remains challenging in practice. First, Condition 1) requires preserving all information relevant to the $R(D^*_\lambda)$-optimal reconstruction, while a strictly bijective decoder is only a restrictive sufficient realization. Second, Condition 2) involves optimization over unrestricted $Q_{\mathbf{Z}|\mathbf{H}}$ and $P_{\mathbf{Z}}$, whereas practical neural parameterizations are confined to limited distribution families and generally yield approximate solutions. Third, Condition 3) requires the conditional expected codelength for each $\mathbf{H}_i$ to match $D_{\mathrm{KL}}\left(Q^{\lambda*}_{\mathbf{Z}|\mathbf{H}_i} \parallel P^{\lambda*}_{\mathbf{Z}}\right)$, which is difficult to achieve with practical coding schemes [31], [34], [35]. Nevertheless, these conditions reveal complementary requirements on information-preserving reconstruction, probabilistic modeling, and coding, thereby providing concrete guidance for the design of near-optimal CSI feedback schemes.

TABLE III
SYMBOL LIST FOR SECTION IV

Notation	Definition
$Q_{\mathbf{Z}\|\mathbf{H}}$	Transition kernel from CSI $\mathbf{H}$ to latent $\mathbf{Z}$
$P_{\mathbf{Z}}$	Prior distribution of $\mathbf{Z}$.
$\mu_{i,j}$	Mean parameter of $Q_{\mathbf{Z}\|\mathbf{H}}$ for the j-th component of the i-th latent block.
$\tilde{\mu}_{i,j}$	Mean parameter of $P_{\mathbf{Z}}$.
$\tilde{\sigma}_{i,j}$	Scale parameter of $P_{\mathbf{Z}}$.
$\sigma_{i,j}$	Scale parameter of $Q_{\mathbf{Z}\|\mathbf{H}}$ when Q and P are set to diagonal Gaussians in the upper-bound evaluation procedure.

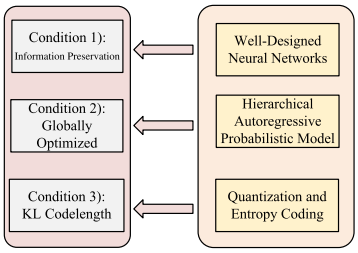


Fig. 2. General strategies of the design of RDFI-Net.

IV. RDFI-NET FOR CSI FEEDBACK

In this section, we present a practical design that builds upon the previously established optimality conditions. Specifically, we develop a VAE-based neural network, termed RDFI-Net, that approximates these conditions for CSI feedback. We first outline the general design principles of RDFI-Net, followed by a detailed description of its architecture. We then introduce a quantization-compatible training strategy, including both fixed-rate and variable-rate modes to enable rate adaptation. Finally, we detail the deployment of RDFI-Net. Key notations used in this section are summarized in Table III.

A. General Strategies

The design of RDFI-Net is based on the optimality conditions established in Proposition 3, with the general strategies illustrated in Fig. 2.

To approximate Condition 1), we aim to prevent the decoder from discarding the information in $\mathbf{Z}$ that is relevant to $\mathbf{H}$. Since preserving all reconstruction-relevant information is difficult to guarantee in practical neural implementations, and strict bijectivity is only a restrictive sufficient condition, we employ an expressive decoder to better approximate the desired reconstruction mapping over the support of the latent

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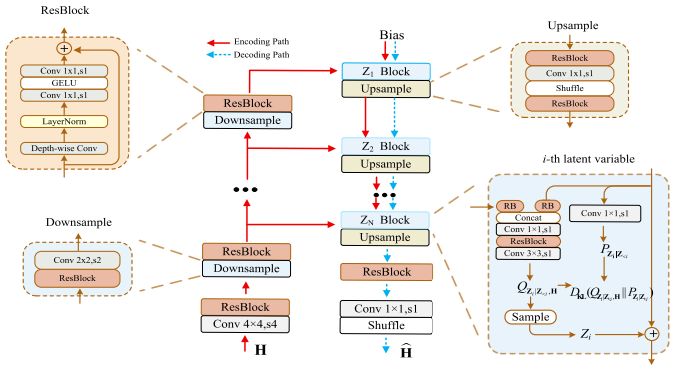


Fig. 3. Architecture of RDFI-Net. Red and blue arrows denote the encoding and decoding paths, respectively. The codec employs hierarchical autoregressive latent variables $\mathbf{Z} \triangleq \{\mathbf{Z}_1, \ldots, \mathbf{Z}_N\}$; the i-th latent block parameterizes the transition kernel $Q_{\mathbf{Z}_i|\mathbf{H},\mathbf{Z}_{<i}}$ and the prior $P_{\mathbf{Z}_i|\mathbf{Z}_{<i}}$ in (22). The "Sample" node indicates the training-time sampling operation, while in deployment each $\mathbf{Z}_i$ is quantized and entropy-coded using the PMF induced by $P_{\mathbf{Z}_i|\mathbf{Z}_{<i}}$ (see Fig. 5–6).

variables. In addition, an invertible pixel-shuffle rearrangement is adopted in the upsampling path [36] to avoid unnecessary information loss during feature reorganization.

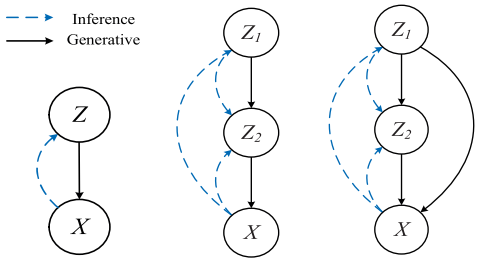
To approximate Condition 2), we employ HAPM [37], [38], [39] as a more expressive parameterization of $Q_{\mathbf{Z}|\mathbf{H}}$ and $P_{\mathbf{Z}}$ in (17). Unlike existing entropy-coded CSI feedback models based on single-layer factorized latents [12], [13], [14] or a sparsity-driven two-tier AE that jointly compresses a sparse latent feature and its side information [15], HAPM conditions each $\mathbf{Z}_i$ on all $\mathbf{Z}_{<i} \triangleq \{\mathbf{Z}_1, \ldots, \mathbf{Z}_{i-1}\}$, explicitly modeling dependencies on all preceding latent blocks. See Section IV-B for details.

To approximate Condition 3), we discretize the latent space $\mathcal{Z}$ through quantization and employ entropy coding to approximate the optimal feedback overhead for $\mathbf{H}_i$, which is given by $D_{\mathrm{KL}}(Q^{\lambda*}_{\mathbf{Z}|\mathbf{H}_i} \parallel P^{\lambda*}_{\mathbf{Z}})$. Under deterministic quantization, the implemented transition kernel reduces to a point mass at $\hat{\mathbf{z}}_i = \mathcal{Q}(f_{\mathrm{en}}(\mathbf{H}_i))$, i.e., $Q^{\mathrm{det}}_{\hat{\mathbf{Z}}|\mathbf{H}_i} = \delta_{\hat{\mathbf{z}}_i}$. Let $P^{\mathrm{PMF}}_{\hat{\mathbf{Z}}}$ denote the discrete PMF induced by the learned prior over the quantization bins. Accordingly,

$$D_{\mathrm{KL}}\left(Q^{\mathrm{det}}_{\hat{\mathbf{Z}}|\mathbf{H}_i} \parallel P^{\mathrm{PMF}}_{\hat{\mathbf{Z}}}\right) = \sum_{z} Q^{\mathrm{det}}_{\hat{\mathbf{Z}}|\mathbf{H}_i}(z) \log_2 \frac{Q^{\mathrm{det}}_{\hat{\mathbf{Z}}|\mathbf{H}_i}(z)}{P^{\mathrm{PMF}}_{\hat{\mathbf{Z}}}(z)} = -\log_2 P^{\mathrm{PMF}}_{\hat{\mathbf{Z}}}(\hat{z}_i). \tag{21}$$

The right-hand side of (21) is the ideal codelength targeted by entropy coding, so deterministic quantization followed by entropy coding provides a practical approximation to Condition 3) [35], [40]. During training, the non-differentiable quantization operation is approximated by additive uniform noise to enable end-to-end optimization, following the common practice in [41].

Our overall design strategy constitutes an RDFI CSI feedback framework: we first formulate CSI feedback from an R–D-theoretic perspective and derive a computable variational upper-bound benchmark, and then use the resulting structural insights to guide practical neural codec design. The present implementation is based on a VAE, while extensions to other probabilistic generative models, such as diffusion models [42], may be explored in future work.



(a) Single layer. (b) Stacking. (c) Autoregressive.

Fig. 4. Comparisons of the probabilistic models in VAE-based CSI feedback schemes.

B. Architecture of RDFI-Net

Based on the general strategies, we herein present the architecture of RDFI-Net, as depicted in Fig. 3. RDFI-Net comprises an encoder and a decoder, corresponding to the encoding path and decoding path in the figure, respectively, and they share the same latent blocks, as a common practice in the hierarchical VAEs [37], [38].

To enhance the probability modeling capacity, RDFI-Net is built based on the HAPM, which contains a set of latent variables $\mathbf{Z} \triangleq \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_N\}$ connected to each other in an autoregressive manner in the sense that the i-th latent variable $\mathbf{Z}_i$ is conditionally dependent on $\mathbf{Z}_{<i} \triangleq \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_{i-1}\}$. Therefore, the transition kernel and prior distribution can be expressed as

$$\begin{aligned} Q_{\mathbf{Z}|\mathbf{H}} &= \prod_{i=1}^{N} Q_{\mathbf{Z}_i|\mathbf{Z}_{<i},\mathbf{H}} = \prod_{i=1}^{N}\prod_{j=1}^{d_i} Q_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i},\mathbf{H}}, \\ P_{\mathbf{Z}} &= \prod_{i=1}^{N} P_{\mathbf{Z}_i|\mathbf{Z}_{<i}} = \prod_{i=1}^{N}\prod_{j=1}^{d_i} P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}, \end{aligned} \tag{22}$$

where $\mathbf{Z}_i^{(j)}$ denotes the j-th component of $\mathbf{Z}_i$ and d_i denotes the dimension of $\mathbf{Z}_i$. Note that the latent block exhibits distinct functionalities in the training, encoding, and decoding phases. For simplicity, Fig. 3 illustrates the latent block in a general form. Their functionalities will be detailed in Sections IV-C and IV-D, respectively.

Fig. 4 illustrates the probabilistic models adopted by representative entropy-coded VAE CSI feedback schemes. Most existing entropy-coded VAE feedback methods rely on a Vanilla VAE architecture, i.e., a single latent layer with a factorized probabilistic model [12], [13], [14], as shown in Fig. 4a. In contrast, HAPM introduces inter-connected latent layers to enhance probabilistic modeling capability [37], [38], [39]. Notably, in contrast to naively stacked latent layers (Fig. 4b) in which $\mathbf{Z}_i$ only depends on $\mathbf{Z}_{i-1}$ and deeper-layer collapse may occur [39], HAPM enables effective cross-layer information fusion through residual skip connections between latent layers, resulting in a more expressive probabilistic model [37], [38], [39]. We also note that the authors in [15] propose a sparsity-driven two-tier design that jointly compresses the

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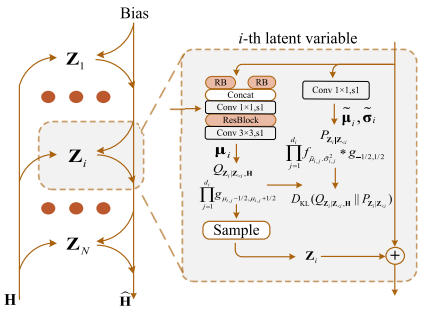


Fig. 5. Illustration of the fixed-rate training strategy of RDFI-Net.

sparse latent feature coefficients and their binary support. Compared with this two-tier feature/support split that hinges on the sparsity assumption, HAPM provides a more general multi-layer latent modeling framework by explicitly capturing inter-layer dependencies, which (to the best of our knowledge) has not been explored for CSI feedback.

Additionally, guided by the information-preservation requirement in Condition 1), the neural backbone of RDFI-Net is designed to be both expressive and information-preserving. Specifically, residual blocks are inserted at each feature-extraction stage, and a ConvNeXt-style design [43] is adopted to strengthen representation learning while maintaining stable optimization. Spatial resolution reduction is implemented via strided convolutions (with stride equal to the kernel size), whereas resolution expansion is performed using sub-pixel convolution (pixel shuffle). Since pixel shuffle is a deterministic and invertible channel-to-space rearrangement, this operation itself preserves information during upsampling and helps avoid artifacts introduced by interpolation or transposed convolutions.

C. Quantization-Compatible Training Strategy

In this subsection, we detail the quantization-compatible training strategy. We first present the fixed-rate training strategy of RDFI-Net. Then, we detail the variable-rate strategy for rate adaptation.

1) Fixed-Rate Training Strategy: Fig. 5 illustrates the fixed-rate training strategy of RDFI-Net. To facilitate end-to-end training that accommodates the non-differentiable nature of quantization, we approximate the quantization operation via additive uniform noise during the training phase, as expressed by

$$\mathbf{Z}_i^{(j)} \leftarrow \mathbf{Z}_i^{(j)} + u, \tag{23}$$

where $\mathbf{Z}_i^{(j)}$ is as in (22) and $u \sim \mathcal{U}(-1/2, 1/2)$. To accommodate this approximation, $Q_{\mathbf{Z}_i|\mathbf{H},\mathbf{Z}_{<i}}$ is selected as a factorized conditional uniform distribution inspired by [41], and the transition kernel is given by

$$\begin{aligned} Q_{\mathbf{Z}|\mathbf{H}} &= \prod_{i=1}^{N}\prod_{j=1}^{d_i} Q_{\mathbf{Z}_i^{(j)}|\mathbf{H},\mathbf{Z}_{<i}}\left(z_{i,j}\right) \\ &= \prod_{i=1}^{N}\prod_{j=1}^{d_i} g_{\mu_{i,j}-\frac{1}{2},\mu_{i,j}+\frac{1}{2}}\left(z_{i,j}\right), \end{aligned} \tag{24}$$

where $z_{i,j}$ denotes the j-th component of $\mathbf{z}_i$, and $\mu_{i,j}$ denotes the mean of $\mathbf{Z}_i^{(j)}$, which is dependent on $\mathbf{H}$ and $\mathbf{Z}_{<i}$. For the selection of $P_{\mathbf{Z}}$, we first show the following lemma.

Lemma 2 (cf. [32]): Let $Q^{\lambda*}_{\mathbf{Z}|\mathbf{H}}$ be the optimal transition kernel that minimizes the objective function in (18). The corresponding optimal prior distribution $P^{\lambda*}_{\mathbf{Z}}$ is given by

$$P^{\lambda*}_{\mathbf{Z}} = \mathbb{E}_{\mathbf{H}\sim P_{\mathbf{H}}}\left[Q^{\lambda*}_{\mathbf{Z}|\mathbf{H}}\right]. \tag{25}$$

Motivated by the lemma above, $P_{\mathbf{Z}}$ should exhibit sufficient flexibility to match the transition kernel. Additionally, $P_{\mathbf{Z}}$ should be strictly positive over the entire support and can be readily converted into a probability mass function (PMF) to facilitate entropy coding. Therefore, $P_{\mathbf{Z}}$ is chosen as

$$\begin{aligned} P_{\mathbf{Z}} &= \prod_{i=1}^{N}\prod_{j=1}^{d_i} P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}\left(z_{i,j}\right) \\ &= \prod_{i=1}^{N}\prod_{j=1}^{d_i}\left(f_{\tilde{\mu}_{i,j},\tilde{\sigma}^2_{i,j}} * g_{-\frac{1}{2},\frac{1}{2}}\right)\left(z_{i,j}\right), \end{aligned} \tag{26}$$

where $P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}$ denotes the PDF of the prior distribution of $\mathbf{Z}_i^{(j)}$. Importantly, $P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}$ can be directly converted into a PMF for entropy coding, as the resulting PMF can be explicitly expressed as

$$\begin{aligned} P^{\mathrm{PMF}}_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}(n) &= \int_{\tilde{\mu}_{i,j}+n-\frac{1}{2}}^{\tilde{\mu}_{i,j}+n+\frac{1}{2}} f_{\tilde{\mu}_{i,j},\tilde{\sigma}^2_{i,j}}(u)\,\mathrm{d}u \\ &= P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}\left(\tilde{\mu}_{i,j}+n\right), \quad n\in\mathbb{Z}. \end{aligned} \tag{27}$$

Equation (27) shows that $P^{\mathrm{PMF}}_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}$ is obtained by evaluating $P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i}}$ at the mean-centered reconstruction levels $\tilde{\mu}_{i,j}+n$, $n\in\mathbb{Z}$, which are spaced by one. Substituting (24) and (26) into (18), the objective function in the fixed-rate training is expressed as

$$\begin{aligned} \mathcal{L}_{\mathrm{FR}}(\boldsymbol{\theta};\lambda) = \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\Bigg[&\sum_{i=1}^{N}\log_2\left(\frac{1}{P_{\mathbf{Z}_i|\mathbf{Z}_{<i}}(\mathbf{Z}_i)}\right) \\ &+\lambda\rho_\omega(\mathbf{H},\mathbf{Z})\Bigg]. \end{aligned} \tag{28}$$

where $\boldsymbol{\theta}$ denotes the parameters of RDFI-Net and λ is as in (18).

Although (28) is algebraically similar to the weighted R–D objectives adopted in existing entropy-coded CSI feedback schemes [12], [13], [14], [15], such objectives primarily serve as operational training criteria for balancing feedback rate and reconstruction distortion. In our formulation, (28) follows from instantiating the variational upper-bound objective in (18) with a quantization-compatible transition kernel and prior. Under deterministic encoding, quantization followed by entropy coding provides one practical realization. More generally, within the RDFI framework, coding schemes whose expected codelength satisfies Condition 3) can be accommodated, with relative entropy coding being one example [34]. This connection provides an R–D-theoretic interpretation of the commonly used weighted R–D objective and links practical neural network training to the variational upper bound $R_\omega(D)$.

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2) Variable-Rate Training Strategy for Rate Adaptation: The optimization of (28) yields a fixed-rate model with a specific rate–distortion trade-off governed by λ, as it specifies an operating point on the R–D function upper bound. Consequently, achieving rate adaptation by changing the operating point would require training multiple distinct models with different λ values, which is resource-intensive and difficult to manage in practice. To enable variable-rate CSI feedback with a single model, we treat λ as a conditioning variable drawn from a predefined distribution P_λ and condition the probabilistic model accordingly. This design avoids training and maintaining multiple models as in previous works [18], [19], thereby reducing the practical burden. Consequently, the objective function for variable-rate training is formulated in (29), shown at the bottom of the page, where $Q_{\mathbf{Z}|\mathbf{H},\lambda}$ and $P_{\mathbf{Z}|\lambda}$ denote the transition kernel and prior distribution conditioned on λ, respectively.

Instead of directly conditioning the entire neural network architecture on λ, which would require architectural modifications or additional conditioning modules, we adopt a low-complexity approach based on a learnable adaptive quantization factor (AQF) motivated by [44] and [45], denoted as $a(\lambda)$. The AQF is coupled with λ and controls the quantization bin size to $\frac{1}{a(\lambda)}$, thereby balancing feedback overhead and reconstruction accuracy. Intuitively, λ is the Lagrange multiplier that selects a desired R–D trade-off. In deployment, AQF offers an implementable control to realize this trade-off by adjusting the effective quantization step $\Delta(\lambda) = \frac{1}{a(\lambda)}$. Specifically, increasing $a(\lambda)$ reduces $\Delta(\lambda)$ and thus refines quantization, typically resulting in a higher bitrate and lower distortion; decreasing $a(\lambda)$ increases $\Delta(\lambda)$ and thus coarsens quantization, typically leading to a lower bitrate and higher distortion. Hence, within the calibrated operating range, adjusting the AQF enables fine-grained adjustment of the empirical R–D operating point and serves as a practical surrogate for varying λ. Consequently, the transition kernel $Q_{\mathbf{Z}|\mathbf{H}}$ can be reformulated to depend on $a(\lambda)$. For notational brevity, let $a \equiv a(\lambda) > 0$ in the following expressions:

$$Q_{\mathbf{Z}|\mathbf{H},a} = \prod_{i=1}^{N} Q_{\mathbf{Z}_i|\mathbf{H},\mathbf{Z}_{<i},a} \tag{31}$$

$$\begin{aligned} Q_{\mathbf{Z}_i|\mathbf{H},\mathbf{Z}_{<i},a}\left(\mathbf{z}_i\right) &= \prod_{j=1}^{d_i} Q_{\mathbf{Z}_i^{(j)}|\mathbf{H},\mathbf{Z}_{<i},a}\left(z_{i,j}\right) \\ &= \prod_{j=1}^{d_i} \left[a g_{\mu_{i,j}-\frac{1}{2a},\mu_{i,j}+\frac{1}{2a}}\left(z_{i,j}\right)\right], \end{aligned} \tag{32}$$

where $\mu_{i,j}$ is as in (24). Similarly, $P_{\mathbf{Z}}$ can be reformulated as

$$P_{\mathbf{Z}|a} = \prod_{i=1}^{N} P_{\mathbf{Z}_i|\mathbf{Z}_{<i},a} \tag{33}$$

$$\begin{aligned} P_{\mathbf{Z}_i|\mathbf{Z}_{<i},a}(\mathbf{z}_i) &= \prod_{j=1}^{d_i} P_{\mathbf{Z}_i^{(j)}|\mathbf{Z}_{<i},a}\left(z_{i,j}\right) \\ &= \prod_{j=1}^{d_i} \left(f_{\tilde{\mu}_{i,j},\tilde{\sigma}_{i,j}^2} * \left[a g_{-\frac{1}{2a},\frac{1}{2a}}\right]\right)\left(z_{i,j}\right), \end{aligned} \tag{34}$$

where $\tilde{\mu}_{i,j}$, $\tilde{\sigma}_{i,j}$ are as in (26). At a quantized reconstruction point $\hat{\mathbf{z}}_i$ satisfying $\hat{z}_{i,j} = \tilde{\mu}_{i,j} + n_{i,j}/a$, $n_{i,j} \in \mathbb{Z}$, with the prior parameters evaluated from the previously decoded latent blocks $\hat{\mathbf{Z}}_{<i}$, the corresponding conditional PMF for entropy coding is given by

$$\begin{aligned} P^{\mathrm{PMF}}_{\hat{\mathbf{Z}}_i|\hat{\mathbf{Z}}_{<i},a}(\hat{\mathbf{z}}_i) &= a^{-d_i} P_{\mathbf{Z}_i|\hat{\mathbf{Z}}_{<i},a}(\hat{\mathbf{z}}_i) \\ &= \prod_{j=1}^{d_i} \int_{\hat{z}_{i,j}-\frac{1}{2a}}^{\hat{z}_{i,j}+\frac{1}{2a}} f_{\tilde{\mu}_{i,j},\tilde{\sigma}_{i,j}^2}(u)\,\mathrm{d}u \\ &= \prod_{j=1}^{d_i} \left(f_{a\tilde{\mu}_{i,j},(a\tilde{\sigma}_{i,j})^2} * g_{-\frac{1}{2},\frac{1}{2}}\right)\left(a\hat{z}_{i,j}\right), \end{aligned} \tag{35}$$

which implies that quantizing $\mathbf{Z}_i^{(j)}$ with bin width equal to $\frac{1}{a}$ is equivalent to first scaling $\mathbf{Z}_i^{(j)}$ with a and then quantizing $a\mathbf{Z}_i^{(j)}$ with a round quantizer. Consequently, the approximation in the training phase is expressed as

$$\mathbf{Z}_i^{(j)} \leftarrow \frac{a\mathbf{Z}_i^{(j)} + u}{a}, \tag{36}$$

where $u \sim \mathcal{U}(-1/2, 1/2)$. The probability distribution of λ is set to

$$P\left(\lambda\right) = \sum_{i=1}^{N_\lambda} \frac{1}{N_\lambda} \delta\left(\lambda - \lambda_i\right), \tag{37}$$

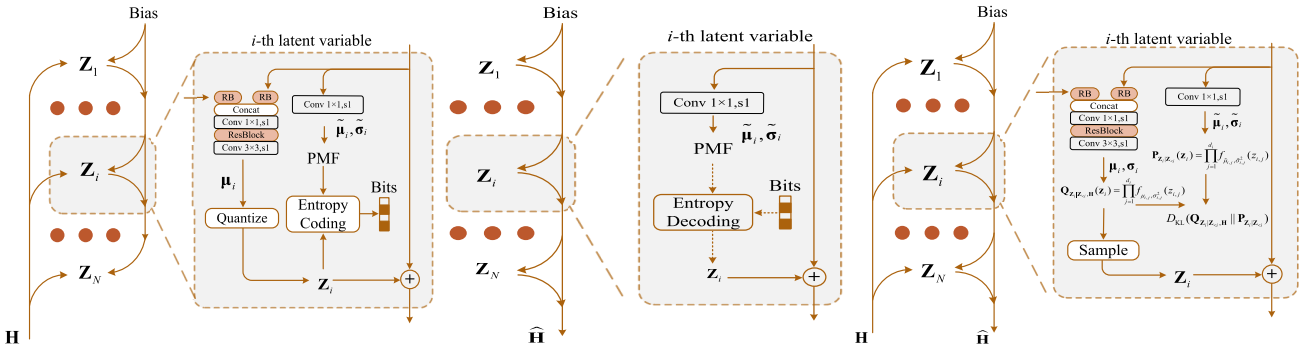
where $\Lambda \triangleq \{\lambda_1, \cdots, \lambda_{N_\lambda}\}$ is a predefined set. Substituting (31), (33) and (37) into (29), the objective function in the variable-rate training is expressed in (30), shown at the bottom of the page, where $\mathcal{A} \triangleq \{a(\lambda_1), \cdots, a(\lambda_{N_\lambda})\}$ denotes the set of AQFs. Optimizing (30) implicitly learns the coupling relationship between λ and $a(\lambda)$.

Under ideal optimization of (30), a larger λ is expected to yield no higher distortion and no lower rate at the predefined operating points. For intermediate AQFs, Λ can be designed to provide denser coverage in the low- and medium-rate region, while adjusting a controls the quantization step through $\Delta(a) = 1/a$ and enables fine-grained rate adjustment between adjacent operating points. As the quantization becomes finer, classical high-resolution quantization approximations [28], [46] further indicate that the rate increases approximately logarithmically with a, whereas the quantization-induced distortion decreases approximately as $1/a^2$. Together, these properties support smooth and approximately monotonic R–D adjustment over the operating range. Compared with training and

$$\mathcal{L}_{\mathrm{VBR}}(\boldsymbol{\theta}) = \mathbb{E}_{\lambda\sim P_\lambda}\left\{\mathbb{E}_{\mathbf{H}\sim P_{\mathbf{H}}}\left[D_{\mathrm{KL}}\left(Q_{\mathbf{Z}|\mathbf{H},\lambda} \parallel P_{\mathbf{Z}|\lambda}\right)\right] + \lambda \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H},\lambda}}\left[\rho_\omega(\mathbf{H},\mathbf{Z})\right]\right\} \tag{29}$$

$$\mathcal{L}_{\mathrm{VBR}}(\boldsymbol{\theta},\mathcal{A}) = \frac{1}{N_\lambda}\sum_{m=1}^{N_\lambda} \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H},a(\lambda_m)}}\left[\sum_{i=1}^{N} \log_2\left(\frac{1}{[a(\lambda_m)]^{-d_i} P_{\mathbf{Z}_i|\mathbf{Z}_{<i},a(\lambda_m)}(\mathbf{Z}_i)}\right) + \lambda_m \rho_\omega(\mathbf{H},\mathbf{Z})\right] \tag{30}$$

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(a) Encoding at UE Side. (b) Decoding at BS Side. (c) Upper Bound Calculation.

Fig. 6. Illustration of the deployment of RDFI-Net.

Algorithm 1 Training of RDFI-Net in Variable-Rate Mode.

Input: Training dataset of downlink CSI $\mathcal{D} = \{\mathbf{H}_i\}_{i=1}^{N_{\rm tr}}$
Output: Trained DNN model parameters $\boldsymbol{\Theta} = \{\boldsymbol{\theta}, \mathcal{A}\}$

Initialize $\boldsymbol{\theta}$ with a pretrained fixed-rate model using a higher λ value, and $\mathcal{A}^{(0)} \triangleq \{a^{(0)}(\lambda_1), \cdots, a^{(0)}(\lambda_{N_\lambda})\}$
Set learning rate η, number of epochs T, batch size B, and load the pretrained model parameters
for epoch = 1 **to** T **do**
  Shuffle the dataset $\mathcal{D}$
  for each mini-batch $\mathcal{B} \subset \mathcal{D}$ of size B **do**
    Sample λ randomly from $\Lambda \triangleq \{\lambda_1, \cdots, \lambda_{N_\lambda}\}$
    Find $a(\lambda)$ from $\mathcal{A}$
    Encoding procedures: $\mathbf{Z} = f_{\boldsymbol{\theta}}^{\rm VBR}(\mathbf{H}, a(\lambda))$
    Approximate quantization with AQF: $\mathbf{Z} \leftarrow \frac{a\mathbf{Z}+\mathbf{u}}{a}$
    Compute the loss $\mathcal{L}_{\rm VBR}$ expressed in (30).
    Compute the gradients: $\nabla_{\boldsymbol{\Theta}} \mathcal{L}_{\rm VBR}$
    Update parameters: $\boldsymbol{\Theta} \leftarrow \boldsymbol{\Theta} - \eta \nabla_{\boldsymbol{\Theta}} \mathcal{L}_{\rm VBR}$
  end for
end for
return: $\boldsymbol{\Theta}$

managing separate models, the proposed mechanism enables a single shared backbone to support fine-grained rate adaptation without being restricted to a predefined set of operating points. For stable optimization, RDFI-Net is first pretrained with a high fixed λ and then fine-tuned using (30), as summarized in Algorithm 1.

D. Deployment

In this subsection, we discuss the deployment of RDFI-Net after the offline training is completed, as illustrated in Fig. 6. The offline training relies on a collection of reference CSI snapshots, which can be acquired during a pilot-based channel sounding/measurement stage (e.g., using longer/denser pilots and averaging across repeated soundings with routine calibration). When the propagation environment changes, periodic calibration data can be used to perform lightweight fine-tuning and update the deployed model.

1) Encoding Procedures: Fig. 6a depicts the encoding of $\mathbf{H}^{\rm BD}$ into binary bits for feedback at UE side. In practice, the encoder of RDFI-Net is deployed at the UE side. The encoder maps $\mathbf{H}^{\rm BD}$ into the latent space, and each $\mathbf{Z}_i$ is obtained by quantizing the residual of $\boldsymbol{\mu}_i$ with respect to $\tilde{\boldsymbol{\mu}}_i$. In fixed-rate mode, this operation is expressed as

$$\mathbf{Z}_i^{(j)} \leftarrow \tilde{\mu}_{i,j} + \mathcal{Q}\left(\mathbf{Z}_i^{(j)} - \tilde{\mu}_{i,j}\right), \quad j = 1, \cdots, d_i. \tag{38}$$

In variable-rate mode, the latent variables are first scaled by the AQF and then quantized, which is expressed as

$$\mathbf{Z}_i^{(j)} \leftarrow \frac{a\tilde{\mu}_{i,j} + \mathcal{Q}\left(a\mathbf{Z}_i^{(j)} - a\tilde{\mu}_{i,j}\right)}{a}, \quad j = 1, \cdots, d_i. \tag{39}$$

After quantization, $\mathbf{Z}_i$ is encoded into binary bits using entropy coding with the corresponding PMF in (27) for fixed-rate mode or (35) for variable-rate mode. Each latent block produces a separate bitstream, and thus the feedback overhead consists of N bitstreams corresponding to $\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_N$.

2) Decoding Procedures: The decoder is deployed at the BS side. When the feedback bit streams are received at the BS side, they are decoded as shown in Fig. 6b. A reconstructed BDCM is obtained at the end of the path and finally SFDCM is reconstructed from the beam based channel model in (7).

3) Upper Bound Evaluation Procedures: Fig. 6c illustrates the calculation of the R–D function upper bound $R_\omega(D)$ when evaluating the variational upper bound $R_\omega(D)$ (as a computable upper-bounding characterization of the R–D limit) with RDFI-Net. For upper bound calculation, $\mathbf{H}^{\rm SF}$ is adopted as the input, and $Q_{\mathbf{Z}_i|\mathbf{H},\mathbf{Z}_{<i}}$ and $P_{\mathbf{Z}_i|\mathbf{Z}_{<i}}$ are set to diagonal Gaussian PDFs. For $j = 1, \ldots, d_i$, their component distributions are parameterized by $(\mu_{i,j}, \sigma_{i,j})$ and $(\tilde{\mu}_{i,j}, \tilde{\sigma}_{i,j})$, respectively. In (40), shown at the bottom of the next page, $D_{\rm KL}$ is measured in bits; its closed-form Gaussian expression is written using natural logarithms and therefore includes the conversion factor $1/\ln 2$. This computed curve serves as a computable variational R–D upper-bound benchmark. Since $R_\omega(D) \geq R(D)$, as shown in Lemma 1, it provides a conservative reference for comparing the achievable R–D performance of practical CSI feedback schemes.

V. Simulations

In this section, we first present the simulation settings. Subsequently, we conduct a series of comprehensive simulations to evaluate the performance of the proposed RDFI-Net.

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TABLE IV
SIMULATION PARAMETERS

Parameters	Values
Outdoor Downlink carrier frequency f_c	3.5 GHz
Indoor Downlink carrier frequency f_c	5.2 GHz
Number of BS antennas N_t	32
Number of subcarriers N_c	512
Number of effective subcarriers N_p	120
Length of cyclic prefix (CP) N_g	72
BS antenna spacing Δ_d	$0.5\lambda_c$
Outdoor channel model	3GPP_38.901_UMa_NLOS; Rician factor: -100 dB
Indoor channel model	3GPP_38.901_Indoor_LOS; Rician factor: $\mathcal{N}(7, 4^2)$ dB
Number of generated clusters	Outdoor: 5; indoor: 5
Subpaths per cluster	Outdoor: 20; indoor: 20
UE speed during QuaDRiGa dataset generation (km/h)	3
Fine factor $F_{\mathrm{an}}, F_{\mathrm{de}}$	2

A. Simulation Settings

To provide a standardized and reproducible evaluation environment aligned with 3GPP NR channel modeling, we generate the downlink space–frequency domain channel matrices using QuaDRiGa v2.4 [47], which implements the 3GPP TR 38.901 geometry-based stochastic channel models. Specifically, we consider the "3GPP_38.901_UMa_NLOS" scenario for outdoor and the "3GPP_38.901_Indoor_LOS" scenario for indoor, with the key parameters summarized in Table IV. To mimic practical CSI acquisition, we discretize the $120\,\mathrm{m} \times 100\,\mathrm{m}$ area into a set of reference points with a $1\,\mathrm{m}$ spacing, resulting in 12,000 CSI samples for each scenario. The dataset is split into $75\%/15\%/10\%$ for training/validation/testing. The corresponding beam-domain channel matrices are obtained via MFOCUSS algorithm [48]. Moreover, to facilitate reproducible benchmarking, we additionally evaluate RDFI-Net on the public COST2100 dataset [4] using the widely adopted split of 100,000/30,000/20,000 samples for training/validation/testing.

The neural network is trained and evaluated using PyTorch 2.4.1 with Python 3.11. All simulations are carried out on a high-performance computing system equipped with an Intel Xeon(R) Gold 6226R CPU and a GeForce RTX 4090 24GB GPU. During training, the total number of epochs is set to 1000, batch size is set to 128 and the learning rate is set to 0.0001. Additionally, a learning rate decay strategy is applied, where the learning rate decays by a factor of 0.9 every 50 epochs. The Adam optimizer is used to update the network's parameters.

During the simulations, we set the number of latent layers of RDFI-Net to 3 and the number of channels for the respective latent layers to $\{16, 32, 64\}$. For variable-rate optimization, we set $\Lambda \triangleq \{0.0018, 0.0035, 0.0068, 0.0131, 0.0255, 0.0495, 0.0960, 0.1862, 0.3611, 0.7006, 1.359\}$, initialize $\mathcal{A} \triangleq \left\{a(\lambda_i) = \sqrt{\lambda_i/\lambda_{N_\lambda}} : i \in \mathcal{Z}^+_{N_\lambda}\right\}$, and learn $\mathcal{A}$ during variable-rate fine-tuning. For variable-rate CSI feedback testing, we vary the AQF from 0.01 to 2, using a step size of 0.05 from 0.01 to 0.9 and a step size of 0.1 from 0.9 to 2. The AQF range is set to cover the target feedback rates considered in the simulations. The entropy codecs in CompressAI [49] are adopted.

To evaluate the feedback performance, we employ three metrics: normalized-mean-square-error (NMSE), cosine distance and bit-per-CSI-unit (BPU). NMSE is defined as

$$\mathrm{NMSE} = \mathbb{E}\left\{\frac{\|\mathbf{H}^{\mathrm{SF}} - \hat{\mathbf{H}}^{\mathrm{SF}}\|_2^2}{\|\mathbf{H}^{\mathrm{SF}}\|_2^2}\right\}, \tag{41}$$

where $||\cdot||_2$ denotes the Euclidean norm. Cosine distance is defined as

$$\eta \triangleq 1 - \mathbb{E}\left\{\frac{\mathrm{Re}\left\{\mathrm{vec}(\mathbf{H}^{\mathrm{SF}})^{\mathrm{H}}\mathrm{vec}(\hat{\mathbf{H}}^{\mathrm{SF}})\right\}}{\|\mathbf{H}^{\mathrm{SF}}\|_2\|\hat{\mathbf{H}}^{\mathrm{SF}}\|_2}\right\}. \tag{42}$$

BPU is defined as $\mathrm{BPU} = \frac{N_{\mathrm{bit}}}{N_t N_p}$, where N_{bit} denotes the total feedback bits. NMSE and cosine distance are used to evaluate reconstruction accuracy while BPU measures the feedback overhead. Unless otherwise stated, NMSE and cosine distance are reported in dB as $10\log_{10}(\mathrm{NMSE})$ and $10\log_{10}(\eta)$, respectively.

On the QuaDRiGa datasets, we compare RDFI-Net with four representative open-source baselines, i.e., CsiNet [4], CLNet [6], CRNet [5], and TransNet [7]. For fairness, these baselines strictly follow their original published architectures (e.g., the number of layers/modules) and implementations, and are retrained from scratch using the same training/validation/testing split and the unified training protocol described above; for each method, the best checkpoint is selected based on the validation NMSE (as the primary criterion), while cosine distance is also monitored and reported at the selected checkpoint. For dimension-reduction-based baselines, each element of the N_{lat}-dimensional latent vector is quantized using 8 bits.[3] Accordingly, the feedback overhead is $N_{\mathrm{bit}} = 8N_{\mathrm{lat}}$, yielding $\mathrm{BPU} = \frac{8N_{\mathrm{lat}}}{N_t N_p}$, where N_{lat} denotes the latent dimension; for entropy-coding-based methods (including

[3]Results reported in DL-based CSI feedback studies indicate that 8-bit quantization is a reasonable practical setting [23], [50].

$$\begin{aligned}\mathcal{L}_{\mathrm{UB}} &= \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\left[\sum_{i=1}^{N} D_{\mathrm{KL}}\left(Q_{\mathbf{Z}_i|\mathbf{H},\mathbf{Z}_{<i}} \parallel P_{\mathbf{Z}_i|\mathbf{Z}_{<i}}\right) + \lambda\rho_\omega(\mathbf{H},\mathbf{Z})\right] \\ &= \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\left[\frac{1}{\ln 2}\sum_{i=1}^{N}\sum_{j=1}^{d_i}\left(-\frac{1}{2} + \ln\frac{\tilde{\sigma}_{i,j}}{\sigma_{i,j}} + \frac{\sigma_{i,j}^2 + (\mu_{i,j} - \tilde{\mu}_{i,j})^2}{2\tilde{\sigma}_{i,j}^2}\right) + \lambda\rho_\omega(\mathbf{H},\mathbf{Z})\right]\end{aligned} \tag{40}$$

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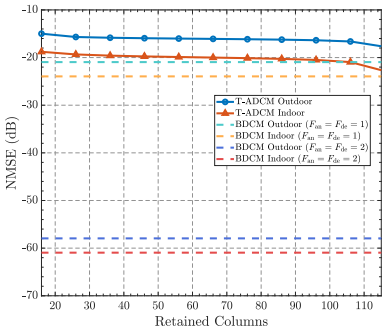
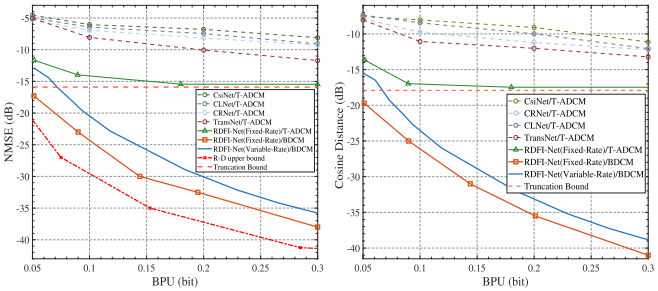


Fig. 7. Reconstruction accuracy vs. retained columns of T-ADCM.



(a) NMSE. (b) Cosine distance.

Fig. 8. Performance on QuaDRiGa Outdoor Dataset.

RDFI-Net), BPU $= \frac{N_{\text{bit}}}{N_t N_p}$ is measured by the actual bit-stream length after entropy coding. We additionally evaluate RDFI-Net on the public COST2100 benchmark [4]. For comparison, all baseline curves are constructed from the COST2100 operating points reported in the corresponding original publications. This avoids uncertainties associated with unofficial reimplementations.

B. Comparison of Reconstruction Accuracy Between T-ADCM and BDCM

Fig. 7 illustrates the variation of reconstruction accuracy as the number of retained columns N_a in the DFT-based ADCM. Here, N_a is varied from N_f to N_p with a step of 5. As shown in the figure, the reconstruction accuracy in both indoor and outdoor scenarios exhibits a plateau, with accuracy improvements becoming marginal as more columns are retained. This phenomenon is primarily due to spectral leakage in the DFT-based ADCM: under the continuous-window interpretation, the finite windowing in the space–frequency domain leads to $\tilde{h}_k^{\text{AD}}(u,\tau) = h_k^{\text{AD}}(u,\tau) *_\tau \Psi_\tau(\tau) *_u \Psi_u(u)$, with $\Psi_\tau(\tau) = N_p\Delta_f \operatorname{sinc}(N_p\Delta_f\,\tau)$ and $\Psi_u(u) = \frac{N_t\Delta_d}{\lambda_c}\operatorname{sinc}\left(\frac{N_t\Delta_d}{\lambda_c}u\right)$, whose sidelobes spread dominant coefficients into the tail columns, as explained in Appendix A. In contrast, with F_{an} and F_{de} both above 1, BDCM achieves superior reconstruction accuracy of -58 dB and -61 dB in outdoor and indoor scenarios, respectively. For comparison, without oversampling (i.e., $F_{\text{an}} = F_{\text{de}} = 1$), the reconstruction NMSE is only about -21 dB (outdoor) and -24 dB (indoor), which is close to the plateau of T-ADCM. This shows the benefits of oversampling. Moreover, BDCM has a dimension of 64×34, which is comparable to that of T-ADCM, indicating that BDCM improves reconstruction accuracy without increasing dimensionality to an unmanageable extent.

C. R–D Performance of CSI Feedback With RDFI-Net

Fig. 8a presents the NMSE performance versus BPU in outdoor scenarios, whereas Fig. 8b shows the corresponding cosine distance performance. In both cases, T-ADCM is used as the input to the baseline models, consistent with their original configurations. For fair comparison, Table V further reports the compression performance of the baseline models with both T-ADCM and BDCM inputs, while the R–D upper bound is computed using SFDCM. The results reveal that the baseline models exhibit limited performance and remain substantially inferior to RDFI-Net over the considered BPU range. In contrast, RDFI-Net with BDCM in fixed-rate mode achieves substantial improvements, attaining an NMSE of -17.04 dB at 0.05 BPU and further improving to -37.8 dB at 0.3 BPU. When using T-ADCM, however, RDFI-Net suffers from a truncation effect, with performance saturating at an NMSE of -15.9 dB and a cosine distance of -18.01 dB, indicating considerable information loss compared with BDCM. From Table V, replacing T-ADCM with BDCM generally improves the baseline performance over the practically relevant range of 0.1–0.3 BPU, with NMSE gains of up to approximately 1.3 dB.

TABLE V
COMPRESSION PERFORMANCE COMPARISON OF T-ADCM AND BDCM WITH BASELINE MODELS

BPU	Methods	T-ADCM		BDCM	
		NMSE	η	NMSE	η
0.05	CsiNet	-4.56	-7.56	-4.01	-7.01
	CLNet	-4.85	-7.36	-4.13	-8.01
	CRNet	-5.30	-7.85	-4.38	-6.72
	TransNet	-5.05	-8.05	-6.25	-9.25
0.1	CsiNet	-6.04	-8.04	-7.22	-9.89
	CLNet	-6.45	-8.45	-7.75	-10.75
	CRNet	-6.95	-9.77	-7.24	-10.15
	TransNet	-8.06	-11.06	-9.33	-12.33
0.2	CsiNet	-6.79	-9.10	-7.61	-10.32
	CLNet	-7.45	-9.98	-8.47	-11.37
	CRNet	-8.16	-11.21	-9.45	-12.80
	TransNet	-10.06	-12.02	-11.21	-14.14
0.3	CsiNet	-8.10	-11.13	-9.16	-12.20
	CLNet	-9.08	-12.05	-9.74	-12.81
	CRNet	-9.27	-12.21	-9.90	-13.48
	TransNet	-11.70	-13.22	-12.05	-14.87

Moreover, RDFI-Net operating in variable-rate mode closely tracks its fixed-rate counterpart, validating its efficiency: although it incurs a slight degradation (about 2 dB in NMSE) because a single shared model must serve multiple target rates and AQF-based scaling cannot fully specialize the representation/entropy model for each operating point, it retains strong feedback performance. The observed smooth and approximately monotonic R–D variation with AQF is consistent with the analysis in Section IV-C.

Finally, since $R_\omega(D) \geq R(D)$, the computed curve provides a conservative upper-bound reference for comparing practical CSI feedback schemes. As shown in the results, RDFI-Net exhibits a smaller gap to this common reference than the evaluated baselines.

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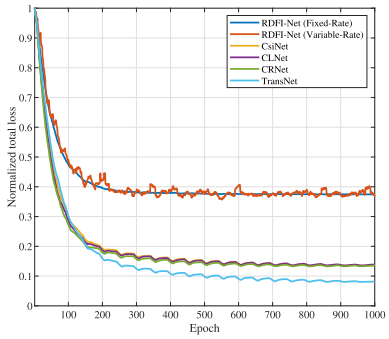
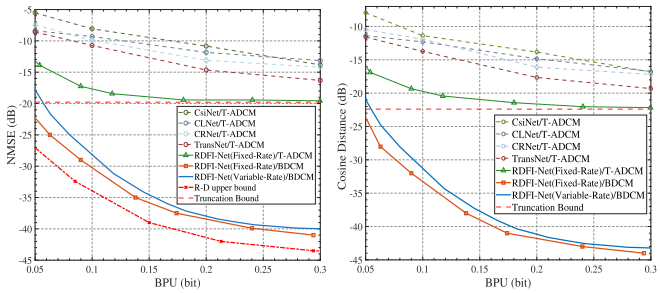
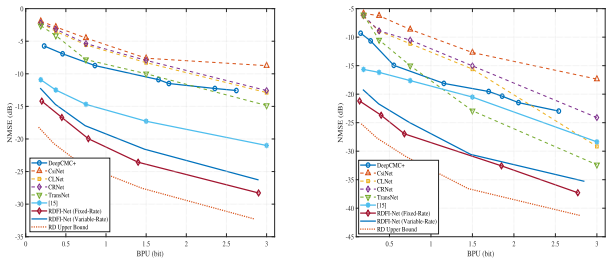


Fig. 9. Training-loss curves on the QuaDRiGa outdoor scenario. The loss values are normalized by the first-epoch loss.



(a) NMSE. (b) Cosine distance.

Fig. 10. Performance on QuaDRiGa Indoor Dataset.



(a) Outdoor. (b) Indoor.

Fig. 11. Performance on the COST2100 dataset. The BPU range of 0–3 is selected to be consistent with the operating range reported in prior COST2100-based studies.

To further examine the stability of AQF-based variable-rate training, Fig. 9 presents the training-loss curves of RDFI-Net. The training loss decreases overall and reaches a stable plateau without divergence or persistent instability, providing empirical evidence of stable training behavior under the adopted protocol.

Fig. 10a and Fig. 10b present the NMSE and cosine distance performance of RDFI-Net in indoor scenarios, respectively. As shown from the results, the T-ADCM-based method again exhibits limited compression performance due to the truncation effect, with NMSE saturating at approximately -18 dB and cosine distance at around -21 dB. In contrast, RDFI-Net with BDCM delivers substantial performance gains, thereby validating the effectiveness of the proposed approach.

To provide a broader public benchmark, we further evaluate RDFI-Net on the COST2100 dataset [4]. We set $\Lambda \triangleq \{0.0360, 0.0700, 0.1360, 0.2620, 0.5100, 0.9900, 1.9200, 3.7240, 7.2220, 14.0120, 27.1800\}$ and otherwise follow the same training protocol as in the QuaDRiGa experiments. For comparison, Fig. 11 also includes the COST2100 results reported for the baseline methods in their original publications. At comparable BPU operating points, RDFI-Net achieves lower NMSE than the reported baselines in both the outdoor and indoor scenarios.

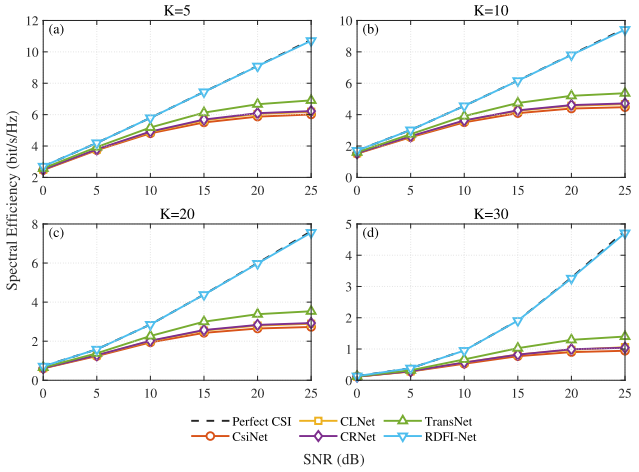
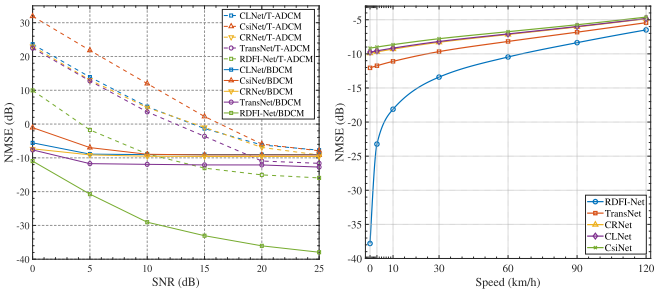


Fig. 12. Per-user spectral efficiency performance.



(a) Imperfect-CSI input perturbations. (b) User mobility.

Fig. 13. Robustness evaluation under complex scenarios.

D. Per-User Spectral Efficiency Performance of CSI Feedback With RDFI-Net

Fig. 12 illustrates the per-user spectral efficiency achieved by different CSI feedback schemes in outdoor scenarios. The number of simultaneously served users is varied as $K \in \{5, 10, 20, 30\}$ to emulate increasingly dense user populations within the same coverage region. For each K, the BS applies zero-forcing precoding based on the reconstructed CSI. All baseline methods and RDFI-Net operate at the same feedback rate of 0.3 BPU, and BDCM is adopted for feedback in all models for a fair comparison. From the results, RDFI-Net consistently outperforms the baselines across all user loads and SNRs. Moreover, even when K increases to 30, RDFI-Net maintains the highest per-user spectral efficiency among the evaluated feedback schemes, demonstrating its relative robustness under dense user deployments.

E. Robustness of CSI Feedback With RDFI-Net

In this subsection, we evaluate the robustness of RDFI-Net under non-ideal conditions. First, we consider controlled input

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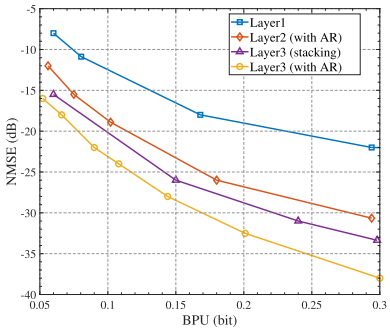


Fig. 14. Ablation simulations on the probabilistic model.

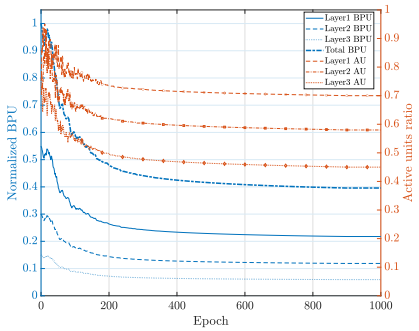


Fig. 15. Training-time layer-wise BPU and AU ratios (fixed-rate training with $\lambda = 1.359$ on the QuaDRiGa outdoor scenario). Total BPU is normalized to 1 for readability.

perturbations that emulate imperfect CSI acquisition. Additive Gaussian noise is introduced into the input channel matrices of the encoders. The Signal-to-Noise Ratio (SNR) is defined as the ratio of the channel matrix power to the Gaussian noise variance. As shown in Fig. 13a, RDFI-Net consistently outperforms baseline models under these controlled input perturbations. Moreover, employing BDCM for CSI feedback not only improves feedback accuracy but also provides greater robustness to input corruption compared to T-ADCM.

Furthermore, we evaluate the performance under the user mobility scenario, with the results shown in Fig. 13b. We generate time-varying CSI snapshots in QuaDRiGa under UE speeds of 0, 3, 10, 30, 60, 90, and 120 km/h. We set a fixed feedback delay of $\Delta t = 1$ ms: the BS reconstructs CSI from an earlier snapshot, while the NMSE is computed with respect to the current CSI after Δt. All models are evaluated under the same feedback overhead of 0.3 BPU. As the UE speed increases, all schemes degrade due to more severe CSI aging, while RDFI-Net consistently achieves the lowest NMSE across the tested speeds.

Fig. 13 can be viewed as partially emulating distribution shifts in practice (e.g., imperfect UE-side CSI acquisition and mobility-induced channel aging), and it shows that RDFI-Net remains robust under such distribution mismatches without retraining. For more significant shifts, the model can be refreshed via lightweight fine-tuning using a small set of newly collected CSI snapshots.

F. Ablation Simulations

Fig. 14 presents ablation results for the HAPM under four outdoor configurations: (a) single-layer, (b) two-layer, (c) a three-layer stacking model, in which each latent block is conditioned only on $\mathbf{Z}_{i-1}$ and the cross-layer skip connections providing $\mathbf{Z}_{<i}$ are removed, and (d) the proposed three-layer HAPM, in which each latent block is conditioned on $\mathbf{Z}_{<i}$. The results show that the single-layer architecture yields the poorest performance, while full-history inter-layer conditioning provides significant compression gains over the stacking baseline. To assess hierarchical latent collapse, Fig. 15 reports the training-time layer-wise BPU and active-unit (AU) ratios. A latent unit is active if the variance of its posterior mean over the training set exceeds 10^{-2} [51]. The AU ratio is the fraction of active units in the corresponding latent layer. All layers retain non-trivial BPU contributions and non-zero AU ratios throughout training, indicating that none collapses entirely; the curves gradually stabilize, suggesting stable optimization.

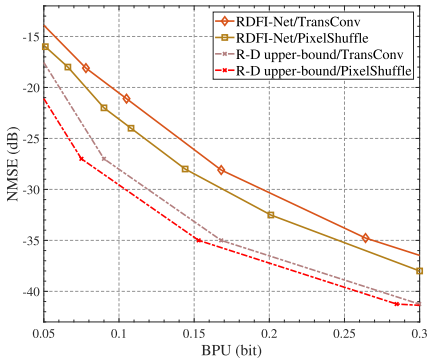


Fig. 16. Ablation simulations on upsampling operations.

TABLE VI
COMPLEXITIES OF DIFFERENT CSI FEEDBACK SCHEMES

Methods	T-ADCM		BDCM	
	Params ($\times 10^3$)	FLOPs (M)	Params ($\times 10^3$)	FLOPs (M)
CsiNet	595.42	7.82	1261.28	16.61
CLNet	594.26	3.90	1260.12	8.29
CRNet	595.14	7.24	1260.99	15.38
TransNet	1171.22	74.94	1837.07	159.24
RDFI-Net	168.63	19.13	168.63	39.27

Fig. 16 reports ablation results for the upsampling modules, comparing (a) transposed convolution and (b) subpixel convolution. The subpixel-convolution design achieves better compression performance in this comparison. Its pixel-shuffle operation is an invertible channel-to-space rearrangement, although the preceding convolution is not necessarily invertible.

G. Complexity

The complexities of RDFI-Net and baseline models are compared in Table VI with the number of learnable parameters (in units of 10^3) and floating-point operations (FLOPs) evaluated. We compare all the models under 0.3 BPU for encoding and decoding. In terms of learnable parameters, the complexity of RDFI-Net is significantly less than those of the baseline models due to its fully convolutional architecture, indicating the potential for deployment. In terms of FLOPs, the

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complexity of RDFI-Net is slightly larger than that of CsiNet, CLNet and CRNet while it is significantly less than that of TransNet. These results indicate a favorable performance–complexity trade-off for RDFI-Net.

VI. Discussion

Recent studies have investigated perception-assisted CSI feedback [52] and superimposed feedback [53], which reduce dedicated feedback overhead through side information or uplink multiplexing under different system assumptions. In contrast, our work follows the standard DL-based CSI feedback setting and focuses on improving compression performance to achieve a better rate–distortion trade-off. These approaches are therefore complementary and can be combined with the proposed framework.

Recent studies have investigated end-to-end CSIT acquisition and derived remote rate–distortion bounds by jointly modeling training and feedback [27], [54], [55]. Our work focuses on CSI compression with UE-available CSI and reliable digital feedback, which is complementary.

VII. Conclusion

In this paper, we investigated downlink CSI feedback for FDD massive MIMO communications with OFDM modulation. By formulating CSI feedback as a lossy compression problem, we studied its fundamental limit governed by the R–D function. To obtain a tractable surrogate for high-dimensional CSI, we employed a variational upper bound that was evaluated through VAE training as a conservative benchmark. Building on this upper bound, we established a set of sufficient optimality conditions that provided theoretical guidance for practical design. Guided by these conditions, we proposed a VAE-based neural network, termed RDFI-Net, for CSI feedback. Specifically, RDFI-Net integrated entropy coding, a hierarchical autoregressive probabilistic model, and enhanced neural modules to approximate the derived optimality conditions. Furthermore, we developed a low-complexity rate adaptation scheme that enabled variable-rate CSI feedback without training and managing multiple models. Numerical results showed that our proposed approach achieved superior performance.

Appendix A
Spectral Leakage in DFT-Based ADCM

Define the angular–delay-domain channel response as

$$h_k^{\mathrm{AD}}(u,\tau)=\sum_{p=1}^{P_k} h_{k,p}\delta(\tau-\tau_{k,p})\delta(u-u_{k,p}), \tag{43}$$

where $P_k, h_{k,p}, u_{k,p}, \tau_{k,p}$ are defined in (2). Therefore, the space-frequency domain channel response is expressed as

$$\begin{aligned} h_k^{\mathrm{SF}}(f,d) &= \int_{-\infty}^{\infty}\int_{-1}^{1} h_k^{\mathrm{AD}}(u,\tau)e^{-j2\pi f\tau}e^{-j2\pi\frac{d}{\lambda_c}u}dud\tau \\ &= \sum_{p=1}^{P_k} h_{k,p}e^{-j2\pi f\tau_{k,p}}e^{-j2\pi\frac{d}{\lambda_c}u_{k,p}}. \end{aligned} \tag{44}$$

Here, f is the baseband frequency offset from f_c. Let d_m and f_n denote the m-th antenna position and the n-th subcarrier offset, respectively. Then $\left[\mathbf{H}_k^{\mathrm{SF}}\right]_{m,n} = \tilde{h}_k^{\mathrm{SF}}(f_n, d_m)$, where $\tilde{h}_k^{\mathrm{SF}}$ is the windowed response, given by

$$\tilde{h}_k^{\mathrm{SF}}(f,d) = h_k^{\mathrm{SF}}(f,d)\, W_f(f)\, W_d(d), \tag{45}$$

with the window functions $W_f(f)=\mathrm{rect}\left(\frac{f}{N_p\Delta_f}\right), W_d(d)=\mathrm{rect}\left(\frac{d}{N_t\Delta_d}\right)$. Multiplication by the window functions in (45) corresponds to convolution with sinc-shaped kernels in the angular–delay domain. Thus, each element of $\mathbf{H}_k^{\mathrm{AD}}$ can be approximated as

$$\left[\mathbf{H}_k^{\mathrm{AD}}\right]_{q,r} \approx \tilde{h}_k^{\mathrm{AD}}(u_q,\tau_r), \tag{46}$$

where $\{u_q\}_{q=1}^{N_t}$ and $\{\tau_r\}_{r=1}^{N_p}$ denote the angular and delay sampling grids with spacings $\Delta_u=\frac{\lambda_c}{N_t\Delta_d}$ and $\Delta_\tau=\frac{1}{N_p\Delta_f}$, respectively. $\tilde{h}_k^{\mathrm{AD}}(u,\tau)$ is given by

$$\begin{aligned} \tilde{h}_k^{\mathrm{AD}}(u,\tau) &\triangleq \mathcal{F}^{-1}_{f,d/\lambda_c}\left\{\tilde{h}_k^{\mathrm{SF}}(f,d)\right\} \\ &= h_k^{\mathrm{AD}}(u,\tau) *_\tau \Psi_\tau(\tau) *_u \Psi_u(u), \end{aligned} \tag{47}$$

where $*_\tau$ and $*_u$ denote convolution with respect to τ and u, respectively, and

$$\begin{aligned} \Psi_\tau(\tau) &= \mathcal{F}_f^{-1}\{W_f(f)\} = N_p\Delta_f\,\mathrm{sinc}(N_p\Delta_f\tau), \\ \Psi_u(u) &= \mathcal{F}^{-1}_{d/\lambda_c}\{W_d(d)\} = \frac{N_t\Delta_d}{\lambda_c}\mathrm{sinc}\left(\frac{N_t\Delta_d}{\lambda_c}u\right). \end{aligned} \tag{48}$$

Here, $\mathrm{sinc}(x) \triangleq \sin(\pi x)/(\pi x)$. The sidelobes of the sinc function lead to non-negligible coefficients in the tail columns of the DFT-based ADCM.

Appendix B
Acquisition of BDCM at UEs

The received signal at the k-th UE, $\mathbf{y}_k$, during downlink pilot transmission is expressed as

$$\begin{aligned} \mathbf{y}_k &= \left(\mathbf{H}_k^{\mathrm{SF}}\right)^T\mathbf{x}_k + \mathbf{z}_k \\ &= \mathbf{U}\left(\mathbf{H}_k^{\mathrm{BD}}\right)^T\mathbf{V}^T\mathbf{x}_k + \mathbf{z}_k, \end{aligned} \tag{49}$$

where $\mathbf{y}_k = [y_k[N_1],\ldots,y_k[N_2]]^T$ and $\mathbf{z}_k = [z_k[N_1],\ldots,z_k[N_2]]^T$ belong to $\mathbb{C}^{N_p\times 1}$, while the subcarrier-independent pilot is $\mathbf{x}_k = [x_k[1],\ldots,x_k[N_t]]^T \in \mathbb{C}^{N_t\times 1}$. Using standard column-wise vectorization, (49) becomes

$$\mathbf{y}_k = \mathbf{A}\mathbf{h}_k^{\mathrm{BD}} + \mathbf{z}_k, \tag{50}$$

where $\mathbf{A} \triangleq \mathbf{U}\otimes\left(\mathbf{x}_k^T\mathbf{V}\right)$ and $\mathbf{h}_k^{\mathrm{BD}} \triangleq \mathrm{vec}(\mathbf{H}_k^{\mathrm{BD}})$. Estimating $\mathbf{h}_k^{\mathrm{BD}}$ from $\mathbf{y}_k$ has been studied in [25].

Appendix C
Proof of Proposition 1

Proof: For convenience, we define

$$R_\omega(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}) \triangleq \mathbb{E}_{\mathbf{H}\sim P_{\mathbf{H}}}\left[D_{\mathrm{KL}}(Q_{\mathbf{Z}|\mathbf{H}} \parallel P_{\mathbf{Z}})\right], \tag{51a}$$

$$D(Q_{\mathbf{Z}|\mathbf{H}}) \triangleq \mathbb{E}_{P_{\mathbf{H}}Q_{\mathbf{Z}|\mathbf{H}}}\left[\rho_\omega(\mathbf{H},\mathbf{Z})\right]. \tag{51b}$$

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It suffices to show that if $Q_{\mathbf{Z}|\mathbf{H}}^{\lambda*}$ and $P_{\mathbf{Z}}^{\lambda*}$ minimize (18), then they also achieve the minimum of R_ω among all $Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}$ such that $D(Q_{\mathbf{Z}|\mathbf{H}}) \le D_\lambda^*$, where D_λ^* is defined in (16b).

For arbitrary $Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}$ satisfying $D(Q_{\mathbf{Z}|\mathbf{H}}) \le \mathcal{D}_\lambda^*$, we have

$$\mathcal{R}_\lambda^* + \lambda \mathcal{D}_\lambda^* \le R_\omega(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}) + \lambda D(Q_{\mathbf{Z}|\mathbf{H}}). \tag{52}$$

Consequently, we have

$$\begin{aligned} \mathcal{R}_\lambda^* &\le R_\omega(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}) + \lambda\big(D(Q_{\mathbf{Z}|\mathbf{H}}) - D_\lambda^*\big) && \text{(53a)} \\ &\le R_\omega(Q_{\mathbf{Z}|\mathbf{H}}, P_{\mathbf{Z}}). && \text{(53b)} \end{aligned}$$

By the definition of $R_\omega(D)$, it follows that $(\mathcal{R}_\lambda^*, \mathcal{D}_\lambda^*)$ lies on $R_\omega(D)$. ■

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